T.C.
BAHÇEŞEHİR UNIVERSITY
GRADUATE SCHOOL
THE DEPARTMENT OF BIG DATA ANALYTICS AND MANAGEMENT
MASTER'S PROGRAM IN BIG DATA ANALYTICS AND MANAGEMENT

TEMPORAL ROBUSTNESS OF MACHINE LEARNING-BASED FLIGHT DELAY PREDICTION: A CASE STUDY AT ISTANBUL AIRPORT

MASTER'S THESIS

SHERAZ LYNA TOUHOUCHE

ISTANBUL 2026

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ABSTRACT

TEMPORAL ROBUSTNESS OF MACHINE LEARNING-BASED FLIGHT DELAY PREDICTION:
A CASE STUDY AT ISTANBUL AIRPORT

Sheraz Lyna Touhouche
Master's Program in Big Data Analytics and Management
Supervisor: Assoc. Prof. Tevfik Aytekin
June 2026, 101 pages

Flight delays remain a major operational challenge in air transportation. Although machine learning has been widely applied to flight delay prediction, most studies rely on random train–test splits that may overestimate real-world performance by ignoring temporal non-stationarity.

This thesis proposes a machine learning framework for predicting binary departure delays (≥15 minutes) at Istanbul Airport using multi-source operational data collected between October 2025 and February 2026. Logistic Regression, Random Forest, and XGBoost models were evaluated using both random-split and temporally consistent validation strategies.

The tuned XGBoost model achieved a ROC-AUC of 0.903 under random-split evaluation but dropped to 0.474 on future operational periods, revealing substantial temporal robustness limitations. Additional robustness experiments improved future-period performance to 0.624 and showed that instability in delay propagation relationships was the primary source of performance degradation.

The findings highlight the importance of temporally consistent evaluation and adaptive machine learning models for reliable deployment in aviation operations.

Keywords: Flight Delay Prediction, Machine Learning, Temporal Robustness, Concept Drift, Delay Propagation

ÖZET

MAKİNE ÖĞRENMESİ TABANLI UÇUŞ GECİKMESİ TAHMİNİNDE
ZAMANSAL DAYANIKLILIK:
İSTANBUL HAVALİMANI VAKA ÇALIŞMASI

Sheraz Lyna Touhouche
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Uçuş gecikmeleri havacılıkta önemli operasyonel sorunlara neden olmaktadır. Makine öğrenmesi tabanlı uçuş gecikmesi tahmin çalışmalarının çoğu, zamansal durağan olmama problemini göz ardı eden rastgele eğitim-test bölmelerine dayanmaktadır.

Bu tezde, Ekim 2025 ile Şubat 2026 arasında toplanan çok kaynaklı operasyonel veriler kullanılarak İstanbul Havalimanı'ndaki ikili kalkış gecikmelerini (15 dakika) tahmin etmeye yönelik bir makine öğrenmesi çerçevesi önerilmiştir. Lojistik Regresyon, Rastgele Orman ve XGBoost modelleri hem rastgele bölme hem de zamansal olarak tutarlı doğrulama yöntemleriyle değerlendirilmiştir.

Ayarlanmış XGBoost modeli rastgele bölmede 0,903 ROC-AUC elde etmiş, ancak gelecek dönem verilerinde performansı 0,474 ROC-AUC'ye düşerek önemli zamansal dayanıklılık sorunlarını ortaya koymuştur. Yapılan dayanıklılık deneyleri geleceğe yönelik performansı 0,624 ROC-AUC'ye yükseltmiş ve performans düşüşünün temel nedeninin gecikme yayılım ilişkilerindeki kararsızlık olduğunu göstermiştir.

Elde edilen bulgular, havacılık operasyonlarında güvenilir uygulamalar için zamansal olarak tutarlı değerlendirme yöntemlerinin ve uyarlanabilir makine öğrenmesi modellerinin önemini ortaya koymaktadır.

Anahtar Kelimeler: Uçuş Gecikmesi Tahmini, Makine Öğrenmesi, Zamansal Dayanıklılık, Kavram Kayması, Gecikme Yayılımı

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Istanbul, June 2026

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LIST OF ABBREVIATIONS

API	Application Programming Interface
AUC	Area Under the Curve
LR	Logistic Regression
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SHAP	SHapley Additive exPlanations
UTC	Coordinated Universal Time
XGBoost	eXtreme Gradient Boosting

Chapter 1

Introduction

Air transportation has become an essential component of modern society, playing a critical role in economic development, tourism, and global connectivity. As global air traffic continues to grow, airports must manage increasingly complex operations under limited infrastructure constraints. Flight delays have emerged as a persistent and costly operational challenge, affecting airlines, airport authorities, and passengers alike. At major hub airports in particular, delays rarely result from a single isolated factor; rather, they arise from the interaction of weather conditions, aircraft rotations, crew constraints, and cascading operational disruptions. Understanding and predicting these delays requires a system-level perspective that accounts for the interconnected nature of airport operations.

Recent advances in data availability and machine learning have opened new possibilities for modeling such complex systems. However, most existing delay prediction studies rely on random train-test splits that artificially intermix temporal periods, likely overestimating real-world performance. The present research addresses this gap through rigorous temporal validation and multi-source data integration at Istanbul Airport.

Istanbul Airport (IATA: IST), which opened in April 2019 following the phased closure of Atatürk Airport, represents one of the most operationally complex hub airports in Europe. Designed to accommodate up to 90 million passengers annually in its initial phase, the airport serves as the primary international hub for Turkish Airlines, which accounts for approximately 51% of total departure traffic. In 2023, Istanbul Airport became the busiest airport in Europe by number of aircraft movements, handling over 620,000 flights and connecting passengers to more than 120 countries (EUROCONTROL, 2024). This concentration of hub operations — in which a single dominant carrier executes dense rotation schedules across shared gates, crew bases, and aircraft fleets — creates conditions in which operational disruptions propagate rapidly through the broader schedule. A single delayed inbound rotation can cascade across multiple subsequent departures within hours, making Istanbul Airport a particularly demanding and consequential environment for delay prediction research.

1.1 Theoretical Framework

This study is grounded in two complementary theoretical traditions. The first is systems theory as applied to air transportation networks, which conceptualises airports as complex adaptive systems in which local disruptions propagate through shared operational resources to produce non-linear, emergent network-wide effects. Within this framework, an individual flight delay is not merely a scheduling failure but a perturbation in a tightly coupled operational system: a delayed inbound aircraft occupies a gate past its scheduled departure time, preventing the subsequent outbound flight from boarding; the inbound crew may simultaneously approach duty-time limits, necessitating reserve crew activation; and connecting passengers from the delayed arrival may trigger holds or misconnections on onward flights. These cascading interactions mean that the delay probability of a given flight is not independent of the operational state of surrounding flights, and predictive models that treat each flight as an isolated observation systematically underestimate the predictability available from network-state features.

The second theoretical tradition is the concept drift literature in machine learning, which recognises that the statistical relationship between input features and a target variable may shift over time as the underlying data-generating process evolves. In operational aviation, concept drift arises from seasonal changes in traffic density, weather regime transitions, airline schedule revisions, and regulatory adjustments that alter the causal structure of delay formation. A model trained on autumn operational patterns learns feature-target relationships that may not generalise to winter operations, even if both periods exhibit similar aggregate delay rates, because the mechanisms driving delays — and the features that most reliably encode those mechanisms — may differ fundamentally across seasonal regimes. Recognition of concept drift has motivated the development of evaluation protocols that test model performance on chronologically separated test periods rather than randomly sampled holdout sets, a methodological principle that the present study applies rigorously.

Together, these two theoretical frameworks provide the conceptual architecture for the present research design. Systems theory motivates the multi-source feature integration strategy, the incorporation of historical delay propagation features, and the delay contagion analysis conducted in Chapter 4. The concept drift perspective motivates the dual evaluation protocol, the temporal stability analysis, and the investigation of whether predictive relationships remain stable across seasonal operational regimes.

1.2 Statement of the Problem

Existing flight delay prediction models remain limited in their ability to accurately represent real-world operational conditions. Traditional approaches rely heavily

on scheduled flight information and basic meteorological variables, treating flights as independent observations and ignoring system-level dynamics. These models implicitly assume that the delay probability of a given flight can be estimated solely from characteristics observable before departure — such as scheduled time, destination, and current weather — without reference to the operational state of the broader airport system. This assumption is violated at high-traffic hub airports, where gate contention, crew positioning requirements, and aircraft rotation schedules create tight interdependencies among flights departing within the same operational window. Ignoring these dependencies means that standard models discard a substantial portion of the predictive information embedded in real-time operational state.

A further limitation concerns evaluation methodology. The predominance of random train-test splits in the published literature artificially intermixes temporal periods, allowing models to be evaluated on historical observations drawn from the same calendar months as their training data. In the presence of seasonal non-stationarity — which is intrinsic to aviation operations — this intermixing produces performance estimates that reflect interpolation within a known distribution rather than genuine forecasting across seasonal boundaries. Published benchmarks, including the widely cited 0.88 ROC-AUC of Gui et al. (2019), were generated under random-split evaluation conditions that do not simulate deployment-realistic forecasting. The degree to which these benchmarks overestimate true operational performance has not been systematically quantified in the existing literature.

At Istanbul Airport specifically, these limitations are particularly consequential. As one of Europe's busiest airports by aircraft movements, Istanbul Airport (IATA: IST, ICAO: LTFM) operates under conditions of extreme hub concentration, with Turkish Airlines executing over 51% of total departures through a schedule organised around dense morning and evening departure banks. This configuration maximises network connectivity but creates severe propagation risk: a disruption to early-morning inbound rotations can cascade through the entire departure schedule before afternoon recovery protocols take effect. Standard prediction models that ignore propagation dynamics, rely on research conducted at lower-congestion airports, or evaluate performance without temporal validation are poorly equipped to inform operational decision-making at IST. The present study directly addresses each of these gaps through a purpose-designed multi-source dataset, a strict temporal validation protocol, and a comprehensive propagation analysis conducted across seasonal periods.

1.3 Purpose of the Study

The purpose of this study is to improve the realism and predictive accuracy of short-term flight delay forecasting at a major international hub airport by integrating

multiple operational data sources and adopting rigorous temporal validation. Rather than focusing solely on scheduled flight and weather data, the proposed approach incorporates historical delay behavior, surface traffic conditions, and airspace congestion. By explicitly testing model performance across seasonal periods, the study also aims to characterize the temporal stability of learned delay patterns and assess the practical deployment readiness of machine learning models for aviation operations.

1.4 Hypotheses/Research questions

This research seeks to answer the following questions:

- **RQ1.** How do machine learning models perform when predicting flight delays at Istanbul Airport using multi-source operational data, and what is the relative influence of different feature categories on predictions?
- **RQ2.** To what extent do predictive relationships learned from one seasonal period generalize to future seasonal periods at a hub airport?
- **RQ3.** What are the mechanisms driving temporal robustness degradation, and which modeling strategies can mitigate it?
- **RQ4.** To what extent does the inclusion of surface road traffic variables improve flight delay prediction performance at Istanbul Airport?

1.5 Significance of the Study

This research contributes to both the flight delay prediction literature and practical aviation operations management through its integration of multi-source operational data and its emphasis on deployment-realistic model evaluation. While many existing studies focus primarily on maximizing predictive performance under conventional random train-test splits, the present study adopts a rigorous temporal validation framework designed to assess whether machine learning models remain reliable when applied to future operational periods. By explicitly examining predictive performance across chronologically separated datasets, the study contributes to ongoing discussions regarding the temporal robustness and real-world applicability of aviation prediction systems.

A second contribution concerns the investigation of delay propagation as a dynamic operational phenomenon within a major hub airport environment. Drawing on systems theory and concept drift perspectives, the study examines whether the relationships linking historical operational conditions to future delays remain stable across seasonal periods. Through the integration of propagation-based features and temporal

robustness analysis, the research provides insight into how evolving operational dynamics influence machine learning model performance and generalizability over time.

The study also contributes to the limited body of research examining the potential relationship between surface road traffic conditions and flight delay outcomes. Although landside accessibility factors are frequently discussed as plausible contributors to airport operations, their predictive value within flight delay prediction models remains largely unexplored. By incorporating traffic-related variables into a comprehensive multi-source prediction framework, the research provides an empirical assessment of their usefulness relative to more traditional operational and meteorological predictors.

From a practical perspective, the findings are intended to support airlines, airport operators, and aviation planners seeking to develop more reliable decision-support systems. The study evaluates the operational implications of different modelling strategies, investigates the stability of predictive relationships across seasonal periods, and examines the role of cost-sensitive decision-making in delay prediction. Collectively, these analyses provide insights that may assist practitioners in designing prediction systems that are not only accurate under historical conditions but also more resilient to changing operational environments.

1.6 Definitions

The following key terms are used throughout this thesis:

Flight delay refers to the difference between the scheduled and actual departure time of a flight. In this study, a departure delay of 15 minutes or more constitutes a significant delay, consistent with international aviation reporting standards (EUROCONTROL CODA; U.S. DOT 14 CFR Part 234).

Delay propagation denotes the phenomenon whereby a delay affecting one flight increases the probability of delays in subsequent flights through shared operational resources such as gates, crew, and aircraft rotations.

Concept drift refers to the change over time in the statistical relationship between input features and the target variable, which causes a trained model's predictive performance to degrade in deployment.

Airspace congestion denotes the level of aircraft density within a defined airspace region surrounding an airport.

Chapter 2

Literature Review

2.1 System-Level Perspectives on Flight Delays

Flight delays represent one of the most persistent and economically consequential challenges in contemporary air transportation. Their effects extend well beyond individual flights, creating cascading inefficiencies that affect airline scheduling, airport capacity utilization, air traffic flow management, and passenger experience. According to industry estimates, flight delays cost the aviation sector tens of billions of dollars annually in direct and indirect costs, spanning fuel burn, crew overtime, aircraft repositioning, and passenger compensation. As global air traffic demand continues to grow, the systemic consequences of congestion and disruption have become increasingly severe, motivating a sustained body of research aimed at understanding and predicting delay dynamics.

Early investigations into flight delays approached the problem from a largely local perspective, examining individual flights or airport pairs in isolation and focusing on scheduled versus actual departure times to identify delay patterns. While these studies produced useful descriptive statistics, they were limited in their ability to explain broader system behavior. Over time, researchers increasingly recognized that flight delays are rarely isolated events. Rather, they emerge from interactions among multiple interconnected subsystems — including runway capacity, gate availability, crew and aircraft rotations, air traffic control flow management, and weather conditions — all operating simultaneously under shared resource constraints.

From a system-level standpoint, air transportation networks exhibit properties commonly associated with complex adaptive systems. Airports, airspace sectors, and flight routes form interconnected networks in which local disruptions propagate spatially and temporally. Congestion at a single hub airport can trigger aircraft rotation delays, crew misconnections, and gate conflicts that ripple through a carrier's entire schedule.Tang et al. (2021) synthesize a broad range of studies adopting a complex network perspective and demonstrate that delay dynamics are strongly influenced by network topology, hub concentration, and congestion intensity. Their review highlights how delays propagate through aviation networks in ways analogous to cascade failures or epidemic processes, underscoring that delay formation is not solely a function of local

conditions but rather an emergent property of the broader system. This perspective is particularly relevant at major hub airports such as Istanbul Airport (IST), where Turkish Airlines' dominant hub operations mean that operational disruptions at a single terminal can produce widespread downstream effects.

The economic consequences of flight delays extend well beyond direct operational costs. According to EUROCONTROL's Central Office for Delay Analysis (2024), flight delays across European airspace generate annual direct costs exceeding 13 billion, encompassing airline fuel overconsumption, crew overtime, and passenger compensation, with indirect costs to passengers and connecting ground transportation adding a further estimated 8–12 billion. At the airport level, delay propagation creates amplifying effects on gate utilisation and turnaround efficiency: a 15-minute ground delay compounding through three downstream rotations can produce an aggregate schedule disruption of 45–60 minutes before airlines activate recovery procedures such as speed-up climb profiles or reduced ground time buffers. These systemic amplification dynamics underscore why individual-flight prediction models that operate without system-level features consistently underperform models that capture the prevailing network state (Tang et al., 2021).

The operational complexity of major hub airports introduces particular challenges for delay prediction research. Hubs concentrate traffic through high-frequency rotations of a dominant carrier, creating dense dependency structures among flights that are largely absent at point-to-point airports. The Turkish aviation market exemplifies this structure: Istanbul Airport processed approximately 76.5 million passengers in 2023, with Turkish Airlines accounting for over 51% of total departures (DHMI, 2024). This carrier concentration creates a fundamentally asymmetric operational environment in which a single carrier's crew scheduling constraints, aircraft rotation patterns, and hub-management decisions dominate aggregate delay dynamics in ways that multi-carrier studies may not capture. Understanding this concentration effect is therefore a prerequisite for building robust single-airport delay prediction systems, and motivates the use of airline-specific historical delay features as primary predictors in the present study.

2.2 Delay Propagation and Historical Delay Features

One of the most influential mechanisms underlying system-level delay behavior is delay propagation — the empirically documented tendency for a delay at one point in time to increase the probability of delays in subsequent periods. This temporal persistence arises through several distinct pathways: delayed aircraft occupying gates beyond their scheduled block time, crews missing positioning connections, passengers misconnecting between inbound and outbound flights, and accumulated congestion

that prevents normal flow recovery. Understanding and encoding this propagation mechanism is widely recognized as essential for accurate predictive modeling.

Empirical research consistently confirms that delays exhibit short-term temporal autocorrelation. The proportion of delayed flights in a recent time window is one of the strongest predictors of current delay probability, reflecting how congestion propagates through a hub's operational cycle. Mamdouh et al. (2023) propose a flight delay prediction framework built around a novel algorithm that constructs features encoding historical delay behavior along specific flight paths. Their approach, which they term the Flight Delay Path Previous-based Machine Learning (FDPP-ML) method, demonstrates that incorporating historical delay context reduces prediction error substantially compared to models trained on static flight characteristics alone. Specifically, their framework achieved an average reduction in mean absolute error of 32% over a 2-hour forecast horizon across a wide network of U.S. airports. Their findings reinforce the view that delay prediction models must account for short-term memory and temporal dependence rather than treating each flight as an independent observation.

A related concern in encoding historical delay features is the risk of temporal data leakage — wherein features computed naively from the training dataset inadvertently incorporate information about the target variable. For example, computing a flight's "previous hour delay rate" by including the flight itself in that calculation creates a circular dependency that inflates training accuracy but collapses under deployment conditions. This methodological issue has received relatively limited attention in the published aviation machine learning literature, despite its practical importance for models intended for operational deployment.

Further evidence on delay propagation comes from studies examining how the effect varies across time-of-day and operational context. Propagation is most severe during peak departure banks — the concentrated wave structures characteristic of hub operations — where gate conflicts and crew misconnections simultaneously affect multiple flights within a narrow scheduling window. Off-peak propagation is substantially lower due to greater schedule slack and reduced resource contention. This temporal heterogeneity implies that effective historical delay features must encode not only the average delay rate in a preceding time window but also the operational context in which those delays occurred. Rolling features aggregated over fixed time horizons conflate high-contagion peak periods with low-contagion off-peak periods, potentially diluting the predictive signal available from congestion state indicators (Mamdouh et al., 2023).

Airline-level heterogeneity in propagation represents a related but under-studied dimension of delay dynamics. Carriers differ substantially in their schedule buffer policies, crew redundancy provisions, and recovery prioritisation procedures, all of which modulate the rate at which an initial delay is absorbed or transmitted to subsequent flights. Network carriers with tighter rotation schedules tend to exhibit higher propa-

gation coefficients than regional or low-cost carriers operating simpler point-to-point networks. Capturing this airline-specific heterogeneity requires carrier-level historical delay aggregations as model features — a design choice that distinguishes the present study's feature engineering from simpler airport-wide aggregation approaches. Empirical quantification of airline-level contagion heterogeneity at a single major hub airport, and its implications for carrier-specific feature engineering, remains largely absent from the published literature — a gap the present study directly addresses.

2.3 Machine Learning Approaches for Flight Delay Prediction

2.3.1 From statistical models to classical machine learning. Quantitative prediction of flight delays has evolved significantly over the past two decades. Early predictive models relied on regression-based techniques and rule-based systems, typically incorporating scheduled departure time, airline identifiers, airport characteristics, and basic meteorological conditions as explanatory variables. These models demonstrated that delays are partially predictable from observable pre-flight information, but their linear functional forms struggled to capture the complex nonlinear interactions among heterogeneous features that characterize aviation operational data.

The increasing availability of large-scale aviation datasets enabled the adoption of more flexible supervised machine learning techniques. Logistic regression, decision trees, and support vector machines have been widely applied to classify flights as delayed or on-time, yielding improvements over purely statistical baselines by allowing nonlinear decision boundaries and higher-dimensional feature representations. Kim et al. (2016) apply a multilevel artificial neural network architecture to predict flight delays at JFK International Airport, leveraging a hierarchical input structure designed to reflect the layered nature of delay causality. Their model outperforms conventional backpropagation networks in terms of prediction accuracy, demonstrating the value of domain-informed architectural choices even in relatively early neural network applications. Similarly, Choi et al. (2016) address weather-induced delay prediction across multiple machine learning algorithms, reporting that model performance is strongly dependent on the quality and temporal resolution of meteorological features, a finding that has since been replicated across numerous applied studies.

An important but frequently overlooked issue in early machine learning applications to delay prediction concerns the implicit stationarity assumption embedded in standard evaluation practices. When a stratified random train-test split is applied to a dataset spanning multiple months, training and test observations are drawn from the same temporal distribution, producing optimistic performance estimates that do not reflect genuine operational forecasting conditions. Regression and classification models evaluated under such conditions may achieve high reported accuracy while

effectively encoding seasonal patterns from the test period that would be unavailable under genuine deployment. The failure to distinguish between interpolation performance — predicting within a known distribution — and extrapolation performance — forecasting genuinely future distributions — has contributed to an optimism bias in the published flight delay prediction literature. Giusti et al. (2022) explicitly document this phenomenon, demonstrating that model performance degrades substantially as the temporal distance between training and test periods increases, and that standard adaptive retraining schedules are insufficient to maintain stable performance across seasons and years. The present study addresses this limitation by evaluating identical models under both random-split and temporally consistent validation protocols in order to assess the impact of evaluation methodology on reported predictive performance.

Despite their improvements over statistical baselines, classical machine learning approaches exhibited limited effectiveness when applied to highly congested or irregular operational conditions, where the interaction effects between temporal, operational, and environmental features become especially complex and where simple feature representations prove inadequate.

2.3.2 Ensemble methods and gradient boosting. More recent applied research has consistently identified ensemble-based machine learning methods as the strongest performers in flight delay prediction tasks. Tree-based ensemble models — particularly Random Forest and gradient boosting variants such as XGBoost and LightGBM — are well suited to structured aviation data characterized by mixed variable types, nonlinear relationships, and complex feature interactions. These methods achieve robustness through averaging or sequential error correction across multiple weak learners, reducing variance without requiring extensive manual feature engineering.

Gui et al. (2019) apply ensemble learning methods to departure delay prediction at Shanghai Pudong International Airport (PVG) and report a ROC-AUC of 0.88, establishing what has become a widely cited performance benchmark for single-airport binary delay classification studies. Their results demonstrate that properly engineered ensemble models can achieve strong discriminative performance even with moderate-sized training datasets, provided that meteorological, operational, and temporal features are comprehensively integrated.

Alfarhood et al. (2024) conduct a comprehensive case study using multi-source operational and meteorological data from Saudi Arabian Airlines, evaluating a wide range of machine learning models including Random Forest, gradient boosting, and neural networks. Their findings confirm that tree-based ensemble and boosting methods consistently outperform linear approaches in terms of discrimination performance. Importantly, their study also adopts an operationally meaningful delay threshold and explicitly discusses the practical relevance of classification-based delay prediction frame-

works for airline operations management.

Mamdouh et al. (2023) further confirm that ensemble approaches benefit significantly from incorporating historical delay context, particularly under congested operational conditions. Taken together, these studies establish ensemble-based gradient boosting — and XGBoost in particular — as the current methodological standard for applied flight delay classification, combining strong predictive performance with computational efficiency and relative interpretability compared to deep learning alternatives.

2.3.3 Deep learning architectures. Deep learning approaches have also been explored for flight delay prediction, particularly in contexts involving complex temporal or spatial dependencies. Recurrent neural networks, including Long Short-Term Memory (LSTM) architectures, have been applied to model sequences of flights and capture long-range temporal dependencies associated with delay propagation. Convolutional neural networks and graph neural networks have been employed to represent spatial relationships among airports and airspace sectors. Yu et al. (2019) investigate deep learning-based delay prediction for commercial air transport at Beijing Capital International Airport (PEK), reporting a ROC-AUC of 0.85 using a deep belief network architecture trained on multi-attribute high-dimensional flight data.

Despite their representational power, the practical advantages of deep learning models in operational delay prediction remain mixed. While deep learning approaches can outperform ensemble methods when very large volumes of high-frequency data are available, they require substantial computational resources, extensive hyperparameter tuning, and large labeled datasets. Moreover, their limited interpretability presents challenges for operational deployment, where transparency and explainability are often critical for gaining practitioner trust and regulatory acceptance. Several recent studies emphasize that model complexity should be aligned with data availability, interpretability requirements, and operational constraints rather than pursued solely for marginal performance gains. For moderate-sized datasets typical of single-airport operational studies — such as the 9,527-flight dataset analyzed in the present research — well-tuned ensemble methods have consistently matched or exceeded deep learning performance while providing substantially greater interpretability.

Consequently, ensemble-based methods remain a strong and widely adopted choice for flight delay prediction tasks involving structured operational data and moderate-sized datasets.

2.4 Feature Engineering and Multi-Source Data Integration

2.4.1 Meteorological features. Weather conditions constitute a major contributor to flight delays and have been extensively studied in the aviation delay prediction

literature. Meteorological factors including wind speed, precipitation, visibility, atmospheric pressure, and ambient temperature can directly affect runway operations, aircraft performance, air traffic control flow management, and safety margins. Numerous studies demonstrate that even relatively coarse weather information, when properly aligned with flight schedules, provides meaningful predictive signal. Adverse weather events tend to amplify delay contagion by simultaneously disrupting multiple flights across a hub, creating conditions in which individual weather measurements capture system-wide operational stress.

The integration of meteorological data into delay prediction models requires careful attention to temporal alignment. Weather observations must be matched to scheduled departure times at appropriate temporal resolutions to avoid information misalignment or inadvertent leakage. Some studies rely on coarse daily weather summaries, while others incorporate hourly or sub-hourly observations. These differences complicate direct comparison of reported performance metrics and raise questions about the robustness of results across varying data granularities. The present research employs hourly weather observations obtained at Istanbul Airport coordinates, using a conservative nearest-neighbor temporal matching strategy with a 4-hour tolerance window to balance coverage and temporal validity.

2.4.2 Temporal and operational features. Beyond direct meteorological effects, temporal patterns play a critical role in delay formation. Delay likelihood varies systematically across hours of the day, days of the week, and seasonal periods, reflecting recurring demand cycles and capacity constraints. Empirical evidence consistently shows that delays cluster during peak traffic windows, reinforcing the importance of circadian and weekly temporal features in predictive modeling. Incorporating short-term historical delay context has been shown to improve predictive stability and reduce sensitivity to transient noise. Operational features such as airline identity, aircraft type, and route-level characteristics further capture systematic differences in operational reliability that persist across time.

The role of temporal features extends beyond simple hour-of-day and day-of-week indicators to encompass multi-scale cyclical patterns that reflect the layered scheduling structure of hub operations. At the daily level, departure banks — the concentrated windows of simultaneous departures characteristic of hub carrier operations — create predictable congestion peaks in which delay probability rises sharply before recovering during inter-bank valleys. At the weekly level, reduced frequencies on weekends translate to lower aggregate delay rates, while holiday periods introduce demand surges that stress capacity constraints in ways that standard temporal features cannot capture. At the seasonal level, demand patterns, weather regimes, and operational protocols introduce cyclical variation in both delay prevalence and propagation intensity. Capturing these multi-scale temporal effects requires feature design that reflects the hierarchi-

cal structure of aviation scheduling rather than treating temporal variables as generic numerical inputs.

Operational identity features — primarily airline carrier indicators — encode systematic differences in punctuality culture, network topology, fleet composition, and hub priority that persist independently of weather and scheduling conditions. At hub airports where a dominant carrier holds priority access to preferred gates and crew resources, carrier identity also proxies for differential access to recovery mechanisms unavailable to smaller operators. One-hot encoded airline dummy variables, despite their high dimensionality, have been shown to carry meaningful discriminative signal in single-airport studies (Alfarhood et al., 2024), reflecting real operational heterogeneity rather than mere label memorisation. The practical implication is that prediction models incorporating carrier identity can distinguish between structurally punctual and structurally delayed carriers, improving calibration across the full distribution of departure events.

2.4.3 Data integration challenges and imputation. A further challenge in multi-source delay prediction research concerns the integration of heterogeneous data sources that differ in temporal resolution, spatial coverage, and completeness. Many published studies rely on proprietary or curated datasets that have been pre-cleaned before analysis, limiting insight into how data quality decisions affect model performance. In practice, operational data frequently contains missing values, timestamp inconsistencies, and encoding errors that require systematic preprocessing. Weather data, in particular, often covers only a fraction of observed flight hours due to API limitations, sensor gaps, or data retention policies.

Several strategies for handling meteorological data missingness have been proposed, ranging from complete-case analysis — which discards observations lacking exact weather matches — to various imputation approaches. The present study implements temporal nearest-neighbor imputation with an explicitly validated tolerance window, supported by autocorrelation analysis demonstrating that adjacent-hour weather observations remain highly correlated ($\rho > 0.73$ at 4-hour lag for all variables). This approach recovers substantially more usable flight records than complete-case analysis while maintaining defensible assumptions about imputation validity.

2.5 Evaluation Methodology and temporal Validity

2.5.1 Limitations of random train-test splits. Evaluation methodology represents a critically important but frequently underexamined dimension of flight delay prediction research. The majority of published studies evaluate model performance using stratified random train-test splits, in which observations are randomly assigned to training and test partitions without regard to their chronological ordering. This

approach reflects standard machine learning evaluation practice but is fundamentally misaligned with operational deployment realities, where models trained on historical data are required to predict outcomes in genuinely future time periods.

Random splits artificially intermix temporal periods, allowing models to train on data from the same calendar month or week as their test observations. In the presence of temporal autocorrelation — which is pervasive in aviation operational data — this produces inflated performance estimates that may substantially overstate real-world deployment effectiveness. Studies employing random splits report performance metrics as though they reflect deployment-realistic conditions, when in fact they reflect a favorable evaluation setting that does not simulate genuine forecasting under distribution shift.

2.5.2 Concept drift and non-stationarity. A closely related challenge concerns temporal non-stationarity in delay patterns — the tendency for the statistical relationships between features and delay outcomes to change over time as a function of seasonal operational dynamics, policy changes, demand fluctuations, and environmental variation. Concept drift, wherein feature-target relationships learned during training no longer hold during deployment, represents a fundamental threat to the real-world effectiveness of static trained models.

Giusti et al. (2022) specifically investigate flight delay prediction under concept drift and demonstrate that model performance degrades substantially as the temporal distance between training and test periods increases. Their study evaluates multiple drift handling strategies and finds that neither static model retraining schedules nor standard adaptive approaches are fully sufficient to maintain stable performance across seasons and years. This finding has significant implications for the operational deployment of aviation delay prediction systems and motivates the investigation of adaptive training strategies, online learning approaches, and season-specific ensemble methods.

More broadly, concept drift has been recognized as a fundamental challenge in machine learning systems deployed in dynamic environments. Unlike static benchmark datasets, real-world operational systems are subject to evolving conditions that may alter both feature distributions and feature-target relationships over time. As a result, models that perform well during development may experience substantial performance degradation when deployed on future data. These concerns are particularly relevant in aviation, where seasonal weather patterns, fluctuating traffic demand, operational policy changes, and airline scheduling adjustments continuously reshape the conditions under which delays occur.

The possibility of seasonal concept drift raises important questions regarding the long-term reliability of flight delay prediction models and the validity of conventional evaluation protocols. These issues motivate the present study's emphasis on temporal validation and the investigation of whether predictive relationships remain stable across

seasonal operational periods.

2.5.3 The case for chronological evaluation. Recognition of these limitations has prompted recent calls for chronologically consistent evaluation protocols in aviation machine learning research, wherein models are trained exclusively on historical data and evaluated on genuinely future periods. Such temporal train-test splits expose distribution shift and pattern instability that are masked by random evaluation designs, providing more honest assessments of deployment readiness. The growing adoption of this evaluation philosophy reflects a broader shift toward operationally realistic benchmarking standards in applied aviation analytics.

The practical implementation of chronological evaluation introduces several methodological decisions that must be handled carefully. The choice of the training-test temporal boundary affects the measured performance gap, as boundaries placed at seasonal transitions may encounter larger distribution shifts than boundaries within a stable seasonal regime. Hyperparameter tuning is equally consequential: using standard k-fold cross-validation on the training set risks selecting configurations that benefit from within-period temporal autocorrelation, requiring instead forward-chaining time-series cross-validation that preserves temporal ordering within folds. The treatment of transitional periods requires deliberate analytical design: including transition months in the training set could partially expose the model to the distribution shift it is intended to detect, while excluding them creates an evaluation window that under-samples the performance trajectory during regime change.

Despite growing recognition of these issues, chronological evaluation remains the exception rather than the rule in published aviation delay prediction research. The majority of studies reviewed in the present chapter employ stratified random splits without discussing the temporal implications of this choice. Where temporal validation is attempted, it is often limited to a single holdout month directly adjacent to the training period, which fails to expose performance under multi-month distributional shift. To the author's knowledge, chronologically separated evaluation designs spanning multiple months remain relatively uncommon within the single-airport flight delay prediction literature. This observation further motivates the need for rigorous temporal validation frameworks capable of assessing model performance under realistic forecasting conditions.

2.6 Model Interpretability and Operational Deployment

Interpretability has emerged as an increasingly critical consideration in aviation machine learning research. While early studies primarily emphasized predictive accuracy, recent work highlights the need to understand model behavior and feature contributions, particularly in safety-critical and operationally sensitive environments

such as commercial aviation, where prediction outputs must be explainable to operations staff, airline management, and regulatory bodies.

Tree-based ensemble models offer inherent interpretability advantages through built-in feature importance measures based on impurity reduction or permutation testing. However, these measures have known limitations, including sensitivity to feature correlation and inability to capture directional effects or interaction-specific contributions. Post-hoc explanation techniques have been developed to address these limitations, with SHAP (SHapley Additive exPlanations) emerging as the leading framework for model-agnostic feature attribution in applied machine learning.

SHAP assigns to each feature a contribution value for each individual prediction, grounded in cooperative game theory's Shapley value framework. This approach has been applied across a range of aviation prediction tasks, providing granular insight into which features drive individual delay predictions and how feature importance varies across operational contexts. Recent studies have demonstrated that SHAP-based analyses reveal operationally meaningful patterns — such as the disproportionate contribution of atmospheric pressure and wind speed to delay probability — that align with domain knowledge about meteorological effects on runway operations and air traffic control procedures. In the context of flight delay prediction, SHAP offers a practical framework for identifying the relative contribution of operational, meteorological, temporal, and historical delay features. Its ability to provide both global and local explanations makes it particularly suitable for operational environments where transparency and interpretability are essential.

Beyond SHAP, several complementary interpretability approaches have been proposed for aviation delay prediction. Partial dependence plots (PDPs) visualise the marginal effect of individual features on predicted delay probability while averaging over the distribution of other features, revealing non-linear response curves that are invisible to linear coefficient analysis. For example, the relationship between wind speed and delay probability typically exhibits a threshold structure: delays remain stable at low-to-moderate wind speeds before rising sharply above operational limits for safe approach and departure. Similarly, temperature shows a non-linear relationship driven by de-icing requirements at low temperatures and convective activity at high temperatures. These threshold effects are well-suited to tree-based models, whose recursive partitioning naturally learns breakpoints in feature space without requiring manual engineering of interaction terms or polynomial transformations.

The operational deployment of interpretable models also carries implications for regulatory compliance and stakeholder trust. Aviation safety regulators and airline operations managers require that automated decision-support systems produce auditable, explainable outputs that can be reviewed, challenged, and overridden by qualified personnel. Black-box deep learning models, regardless of their predictive accuracy, face

significant adoption barriers in safety-critical environments where unexplained predictions cannot be accepted as the basis for resource allocation decisions. The combination of strong predictive performance and SHAP-based interpretability offered by gradient boosting methods such as XGBoost therefore represents a practical optimum for operational aviation deployment — balancing accuracy, computational efficiency, and the transparency requirements of the regulatory environment (Alfarhood et al., 2024).

2.7 Research Gaps and Positioning of the Present Study

Despite substantial progress in the flight delay prediction literature, several important limitations remain. First, the majority of published studies employ random train-test splits that may overestimate real-world deployment performance by masking temporal non-stationarity. Performance benchmarks reported in the literature — including widely cited results from Gui et al. (2019) and Yu et al. (2019) — are typically derived under random evaluation conditions that do not fully replicate genuine forecasting scenarios. As a result, there remains limited evidence regarding how well flight delay prediction models generalize across temporally separated operational periods.

Second, delay propagation analysis in existing studies has largely focused on aggregate or cross-sectional relationships, with relatively limited attention devoted to how propagation dynamics may evolve across different seasonal or operational regimes. The possibility that delay contagion mechanisms vary over time remains insufficiently explored within the single-airport flight delay prediction literature.

Third, despite theoretical arguments linking landside accessibility to airport operational performance, the potential contribution of surface road traffic conditions to flight delay prediction has received relatively little empirical attention. The predictive value of traffic-related variables therefore remains an open question requiring further investigation.

Fourth, many existing studies either rely on simplified feature sets that do not fully capture system-level operational interactions or utilize proprietary datasets that limit reproducibility and independent validation. The growing availability of public flight, weather, and transportation data sources presents new opportunities for developing transparent and reproducible machine learning frameworks for aviation analytics.

The present study seeks to address these limitations through the integration of multi-source operational data, the implementation of a chronologically consistent evaluation framework, and the investigation of temporal robustness in flight delay prediction at a major international hub airport. Particular attention is devoted to the role of delay propagation, the stability of predictive relationships across seasonal periods, the potential contribution of surface traffic variables, and the challenges associated with deploying machine learning models in environments characterized by temporal

non-stationarity. In doing so, the study aims to contribute both methodological and empirical insights to the growing body of research on aviation delay prediction and operational decision support.

Chapter 3

Methodology

3.1 Research Design

This study employs a quantitative research approach using supervised machine learning to predict binary flight delay outcomes at Istanbul Airport. The research design encompasses four primary phases: data collection and integration, feature engineering with temporal leakage prevention, model development and training, and comprehensive evaluation through both random-split and temporal validation strategies.

The methodological framework prioritizes temporal validity and operational realism. Unlike studies employing random train-test splits that artificially intermix temporal periods, this research implements strict chronological separation to assess true predictive performance under deployment conditions. This design choice enables detection of temporal instability in delay patterns—a critical consideration for operational aviation systems where models must generalize to future unseen periods.

The research adopts an iterative refinement approach, developing the dataset and models in two distinct phases. Phase 1 (Dataset V1) establishes baseline methodology with conservative direct weather matching, while Phase 2 (Dataset V1 Imputed Fixed) implements weather imputation and temporal expansion to produce a substantially larger analytical dataset. This iterative process demonstrates scientific rigor through systematic problem identification and resolution.

3.2 Dataset Description

3.2.1 Phase 1 dataset (dataset V1). The initial dataset (Dataset V1) comprised 4,330 flights operating at Istanbul Airport (IATA: IST, ICAO: LTFM) during October and November 2025. This dataset integrated three primary data sources: flight operational records, meteorological observations, and surface traffic measurements, using conservative direct temporal matching without imputation.

Data sources

Flight operational data Flight records were collected via the AviationStack API (https://aviationstack.com), which aggregates real-time and historical flight information

from multiple aviation data providers. The API provides structured JSON responses containing:

- Flight identification (IATA code, airline, route)
- Scheduled departure timestamp (UTC)
- Actual departure timestamp (UTC) when available
- Delay duration (minutes) computed from timestamp difference

Data collection period: October 27, 2025 – November 30, 2025 (35 days)
Raw records retrieved: 42,623 flight records
Records with actual departure times: 9,814 (23.0%)

The 77% missingness rate primarily reflects future scheduled flights lacking actual departure information, cancelled operations, and API data gaps. Exploratory comparisons across observable characteristics such as hour of day, airline, and destination did not reveal obvious systematic patterns in missingness. However, no formal statistical test of the MCAR assumption was conducted. Consequently, the assumption should be interpreted as a practical working assumption rather than a formally verified property of the dataset.

Meteorological data Weather observations were obtained from the OpenWeather API (https://openweathermap.org/api) using Istanbul Airport coordinates (41.2753°N, 28.7519°E). Variables collected include:

- Temperature (°C)
- Relative humidity (%)
- Atmospheric pressure (hPa)
- Wind speed (m/s)
- Weather category (Clear, Clouds, Rain, Snow)

Temporal resolution: Hourly observations
Phase 1 Coverage: 377 hourly records (October: 44h, November: 201h, December: 8h, January: 8h)
Temporal Density: 22% coverage during study period

Phase 1 methodology employed strict direct matching: flights were included only if weather observations existed for their exact scheduled departure hour. This conservative approach ensured data quality but limited dataset size substantially.

Surface traffic data Road traffic indicators were sourced from Istanbul Metropolitan Municipality (İBB) Traffic Management Center, covering major airport access routes (E80/TEM highway, O-3/O-7 ring roads). Due to data access constraints, a representative 24-hour traffic profile was constructed by averaging conditions across each hour-of-day over a 6-month historical window. Three composite indicators were derived:

- `road_avg_speed` (km/h): Mean vehicle speed on monitored routes
- `road_vehicle_count`: Total vehicles per hour
- `road_congestion_index`: Composite metric combining speed reduction and volume relative to free-flow capacity

This approach captures typical circadian traffic patterns (morning/evening rush hours, off-peak periods) but cannot reflect day-to-day variability from weather, accidents, or special events.

Dataset V1 characteristics

Table 1
Dataset V1 Summary Statistics

Characteristic	**Value**
Total flights	4,330
Temporal span	Oct 27 – Nov 30, 2025 (35 days)
Weather observations (hourly)	377 hours
Weather coverage rate	45.4%
Monthly Distribution:	
October 2025	1,435 flights (51.6% delayed)
November 2025	2,887 flights (58.3% delayed)
December 2025	8 flights
Overall delay rate (≥15 min)	56.2%
Features per observation	20 base + airline dummies
Total dimensionality	~461 features

The 45.4% weather coverage rate reflects the conservative constraint that flights could only be included if weather observations existed for their exact scheduled departure hour. This strategy ensured data quality but substantially limited dataset size.

3.2.2 Phase 2 dataset (Dataset V1 imputed fixed). Following identification of severe data coverage limitations in Phase 1, the dataset was reconstructed with temporal weather imputation and extended temporal coverage, yielding Dataset V1 Imputed Fixed comprising 9,527 flights spanning five months.

Methodological improvements

Phase 2 dataset construction implemented four key improvements over Phase 1:

1. **Weather Imputation**: Rather than excluding flights lacking exact weather matches, missing meteorological data were imputed using temporal nearest-neighbor matching within conservative 4-hour windows (detailed in Section 3.3.4). This increased weather coverage from 45.4% to 68.2%.

2. **Extended Temporal Coverage**: Inclusion of December 2025 data (565 flights), providing critical winter operational context and enabling assessment of seasonal delay pattern variation.

3. **Bias Correction**: Removal of inadvertent filtering steps that preferentially excluded certain flight categories, improving representativeness.

4. **Conservative Tolerance**: Weather imputation employed 4-hour tolerance window (rather than 6-hour initially explored), balancing coverage gains against temporal validity concerns.

Dataset V1 imputed fixed characteristics

Table 2
Dataset V1 Imputed Fixed Summary Statistics

Characteristic	**Value**
Total flights	9,527
Temporal span	Oct 2025 – Feb 2026 (5 months)
Weather coverage (post-imputation)	68.2%
Weather imputation tolerance	4 hours
Monthly Distribution:	
October 2025	5,329 flights (79.2% delayed)
November 2025	3,200 flights (79.9% delayed)
December 2025	565 flights (78.4% delayed)
January 2026	241 flights (76.8% delayed)
February 2026	192 flights (77.6% delayed)
Overall delay rate (≥15 min)	79.4%
Delayed flights	7,563
On-time flights	1,964

Dataset Expansion: Phase 2 dataset is 2.2× larger than Phase 1 (9,527 vs. 4,330 flights), providing substantially improved statistical power and enabling more robust model training.

Train-Test Split: Chronological split with October-November 2025 as training data (8,529 flights, 89.5%) and January-February 2026 as test data (433 flights, 4.5%). December 2025 (565 flights, 6.0%) reserved for intermediate temporal validation.

Delay Rate Increase: Phase 2 exhibits higher delay rate (79.4% vs. 56.2% in Phase 1). This reflects three factors: (1) improved data completeness capturing more operational reality, (2) inclusion of winter months (December-February) with elevated delay rates, and (3) correction of selection biases that preferentially excluded delayed flights in Phase 1.

3.2.3 Variables and target definition.

Target variable

The dependent variable is binary delay status, defined as:

$$y = \begin{cases} 1 & \text{if } \delta \geq 15 \text{ minutes (delayed)} \\ 0 & \text{if } \delta < 15 \text{ minutes (on-time)} \end{cases} \tag{3.1}$$

where $\delta = t_{\text{actual}} - t_{\text{scheduled}}$ represents departure delay in minutes.

The 15-minute threshold aligns with international aviation standards established by the U.S. Department of Transportation (14 CFR Part 234), EUROCONTROL's Central Office for Delay Analysis (CODA), and IATA operational reporting guidelines. This threshold distinguishes minor schedule deviations absorbed by operational buffers from significant delays requiring resource interventions (gate reassignments, crew adjustments, passenger connection management).

Predictor variables

Predictor variables span four conceptual categories:

Temporal features

- `hour_of_day`: Scheduled departure hour (0-23), capturing circadian operational patterns
- `day_of_week`: Day identifier (0-6), differentiating weekday vs. weekend dynamics

Meteorological Features

- `temperature`: Ambient air temperature (°C)
- `humidity`: Relative humidity (%)

- `pressure`: Atmospheric pressure (hPa)
- `wind_speed`: Sustained wind speed (m/s)
- `weather`: Categorical condition (Clear, Clouds, Rain, Snow)

Historical Delay Features

- `prev_hour_delay_rate`: Proportion of delayed flights in previous hour
- `avg_delay_last_24h`: Rolling 24-hour average delay rate
- `airline_7d_delay_rate`: Carrier-specific 7-day rolling delay rate

Surface Traffic Features

- `road_avg_speed`: Average vehicle speed on airport access routes (km/h)
- `road_vehicle_count`: Hourly vehicle volume
- `road_congestion_index`: Composite congestion severity metric

Airline Identity

- `airline_name`: Categorical carrier identifier (one-hot encoded)

3.3 Data Preprocessing and Feature Engineering

3.3.1 Data Cleaning.

Missing data handling

Flight records lacking actual departure timestamps were excluded, as the target variable cannot be computed without both scheduled and actual departure times. This resulted in 77% data reduction from raw API records (42,623 $\rightarrow$ 9,814 flights with computable delays).

Meteorological data exhibited 78% missingness during the study period. Phase 1 excluded flights with missing weather (reducing to 4,330 flights), while Phase 2 employed temporal imputation (Section 3.3.4) to recover coverage.

Airline names contained occasional parsing errors manifesting as timestamp values. These erroneous entries (n=47, 0.5%) were manually corrected through cross-referencing with flight IATA codes.

Outlier removal

Delay Outliers Flights exhibiting extreme delays ($|\delta| > 720$ minutes, i.e., >12 hours) were excluded (n=82, 0.8% of valid records). Such cases likely represent:

- Data entry errors (timestamp typos)
- Flight diversions (arrival airport changed, invalidating original schedule)
- Multi-day delays from extraordinary circumstances

These events constitute fundamentally different operational regimes unsuitable for standard delay prediction models.

Weather Variable Outliers Applied three-sigma rule combined with physical plausibility constraints:

$$\text{Outlier if: } |x - \mu| > 3\sigma \quad \text{AND} \quad x \notin [\min_{\text{physical}}, \max_{\text{physical}}] \tag{3.2}$$

Table 3
Weather Variable Outlier Detection

Variable	Physical Range	Outliers	Action
Temperature	[-20°C, 50°C]	3	Removed
Humidity	[0%, 100%]	0	None
Pressure	[980 hPa, 1050 hPa]	2	Removed
Wind Speed	[0 m/s, 25 m/s]	8	Capped at 25 m/s

Total weather outliers removed: 13 observations (1.6% of hourly records).

Timestamp standardization

All temporal data were standardized to Coordinated Universal Time (UTC) using `pandas.to_datetime(utc=True)`. Approximately 3% of records provided timestamps in local time (UTC+3, Turkey Time), identified through cross-validation with known daylight patterns and corrected algorithmically.

3.3.2 Feature selection. Feature selection employed domain-informed inclusion based on aviation operational theory and prior empirical literature, rather than automated statistical selection procedures. This approach ensures features encode aviation-relevant causal mechanisms.

Included Features: All temporal, meteorological, historical, traffic, and airline features listed in Section 3.2.3 were retained, as each encodes plausible delay-causal pathways documented in aviation operations research.

Excluded Features: Destination airport codes were excluded due to extreme cardinality (500+ unique destinations) creating sparse one-hot representations. Route-level analysis was deferred to future work.

3.3.3 Temporal leakage prevention. Temporal data leakage—wherein predictive features inadvertently contain information from or about the target variable—poses a critical threat to model validity in time-series forecasting. This study implements rigorous leakage prevention protocols for historical delay features.

Leakage risk in historical features

Historical delay aggregations (`prev_hour_delay_rate`, `avg_delay_last_24h`, `airline_7d_delay_rate`) summarize past delay patterns. Naive computation including the current flight introduces circular dependency:

$$\texttt{prev_hour_delay_rate}_{\text{naive}} = \frac{\sum_{i \in \text{FlightsInHour}(t)} \mathbb{1}(y_i = 1)}{|\text{FlightsInHour}(t)|} \tag{3.3}$$

This formulation allows the model to "see" whether the current flight is delayed, creating spuriously high predictive performance that collapses under deployment.

Strict temporal ordering protocol

To prevent leakage, historical features for flight i at time t are computed using *exclusively* flights with scheduled departure $< t$:

$$\texttt{prev_hour_delay_rate}_{\text{correct}}(t) = \frac{\sum_{j \in \text{Flights}(t-1)} \mathbb{1}(y_j = 1)}{|\text{Flights}(t-1)|} \tag{3.4}$$

where $\text{Flights}(t-1)$ denotes flights departing in the hour strictly *before* t.

Similarly, rolling averages employ only temporally preceding data:

$$\texttt{avg_delay_last_24h}(t) = \frac{\sum_{j \in \text{Flights}([t-24,t))} \mathbb{1}(y_j = 1)}{|\text{Flights}([t-24,t))|} \tag{3.5}$$

Implementation via chronological iteration

Enforcement required chronological iteration through the dataset rather than vectorized computation:

Listing 3.1: Leakage-free historical feature computation

```
df_sorted = df.sort_values('scheduled_departure_time')

for idx, current_flight in df_sorted.iterrows():
    current_time = current_flight['scheduled_departure_time']
```

```
# Previous hour flights: strictly before current time
prev_hour_flights = df_sorted[
    (df_sorted['scheduled_departure_time'] >= current_time - pd
    (df_sorted['scheduled_departure_time'] < current_time)
]

if len(prev_hour_flights) > 0:
    df_sorted.at[idx, 'prev_hour_delay_rate'] = prev_hour_fligh
```

Leakage detection and validation

Statistical diagnostic Leakage manifests as artificially smooth historical features exhibiting means and variances suspiciously matching the overall delay rate. Post-correction validation confirmed proper implementation:

Pre-correction: Feature mean exactly matches delay rate, indicating leakage.

Post-correction: Feature mean differs substantially from overall delay rate, and variance increases, reflecting genuine temporal variation rather than target contamination.

3.3.4 Feature construction.

Weather data imputation

Meteorological observations covered only 22% of flight hours, necessitating imputation to avoid massive data loss. The imputation strategy balances coverage and quality through temporal nearest-neighbor matching with conservative tolerance constraints.

Methodology For each flight at hour h lacking exact weather observation:

$$h_{\text{weather}} = \arg \min_{h' \in \mathcal{H}_{\text{available}}} |h' - h| \tag{3.6}$$

subject to proximity constraint:

$$|h_{\text{weather}} - h| \leq 4 \text{ hours} \tag{3.7}$$

If no observation exists within the 4-hour window, the flight is excluded from the dataset.

Tolerance Selection Rationale Initial exploration tested 6-hour tolerance. Final methodology deliberately employs 4-hour tolerance, prioritizing data quality over sample size. Meteorological autocorrelation analysis justifies this choice:

Table 4
Weather Variable Temporal Autocorrelation

Variable	1-hour lag	4-hour lag	6-hour lag
Temperature	0.98	0.91	0.87
Humidity	0.94	0.84	0.79
Pressure	0.99	0.95	0.92
Wind Speed	0.86	0.73	0.64
Weather Category	0.91	0.82	0.75

High autocorrelation at 4-hour lag ($\rho > 0.73$ for all variables) validates the assumption that nearby observations provide reasonable proxies.

Imputation quality assessment Leave-one-out cross-validation on exact-match subset quantified imputation accuracy:

Table 5
Weather Imputation Error Analysis

Variable	MAE	Range	Relative Error
Temperature	1.4°C	37°C	3.8%
Humidity	5.1%	74%	6.9%
Pressure	0.9 hPa	31 hPa	2.9%
Wind Speed	1.3 m/s	18 m/s	7.2%
Weather Category	11% mismatch	N/A	N/A

Relative errors remain below 8% for continuous variables, supporting imputation validity.

Categorical encoding

One-Hot encoding Categorical variables (airline name, weather condition) were transformed via one-hot encoding, creating binary indicator features for each category. Airlines with fewer than 30 flights were aggregated into an ”Other” category to prevent overfitting on rare carriers.

Final Feature Dimensionality After encoding:

- Numeric features: 12
- Weather categories: 4 (Clear, Clouds, Rain, Snow)

- Airline categories: 80-120 (varies by dataset split)
- Total features: $p \approx 96$ to 136

3.4 Modeling Approach

The three algorithms selected — Logistic Regression, Random Forest, and XGBoost span the interpretability-complexity spectrum and collectively satisfy the requirements of this study. Logistic Regression provides a linear, fully interpretable baseline whose coefficients allow direct verification of feature-delay relationships against domain expectations. Random Forest and XGBoost were preferred over other gradient boosting frameworks such as LightGBM because their wider adoption in the aviation delay prediction literature enables direct performance benchmarking against published studies. Support vector machines were excluded due to their poor scaling behaviour with high-dimensional one-hot encoded features and their lack of native probability calibration. Neural network architectures were excluded for three reasons: the dataset size (9,527 observations) is insufficient to exploit deep learning's capacity advantages; their limited interpretability conflicts with the study's emphasis on SHAP-based feature attribution for operational deployment; and prior work on comparable single-airport datasets has consistently found that well-tuned ensemble methods match or exceed deep learning performance when training data volume is limited (see Section 2.3.3). This selection therefore prioritises operational deployability and interpretability alongside predictive performance.

3.4.1 Logistic regression. Binary logistic regression models the log-odds of delay as a linear combination of features:

$$\log\left(\frac{P(y=1|X)}{1-P(y=1|X)}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j \tag{3.8}$$

Rationale Provides interpretable baseline with transparent feature-delay relationships via coefficients β_j. Linear decision boundaries enable rapid inference and straightforward deployment.

Implementation `sklearn.linear_model.LogisticRegression` with L2 regularization (Ridge penalty). Regularization strength C selected via 5-fold time-series cross-validation.

Class imbalance handling `class_weight='balanced'` parameter automatically adjusts class weights inversely proportional to class frequencies:

$$w_{\text{class}} = \frac{n_{\text{total}}}{n_{\text{classes}} \times n_{\text{class}}} \tag{3.9}$$

3.4.2 Random forest models. Random Forest constructs an ensemble of decorrelated decision trees via bootstrap aggregation and random feature subsampling:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(X) \tag{3.10}$$

where B is the number of trees and T_b is the b-th bootstrapped tree.

Rationale Captures non-linear feature interactions and threshold effects without manual feature engineering. Robust to overfitting through ensemble averaging and feature randomization.

Implementation `sklearn.ensemble.RandomForestClassifier`

Hyperparameter Tuning Randomized search cross-validation (30 trials, 5-fold time-series CV):

Table 6
Random Forest Hyperparameter Search Space

Parameter	Search Space
`n_estimators`	[100, 200, 300, 500]
`max_depth`	[None, 10, 20, 30, 50]
`min_samples_split`	[2, 5, 10]
`min_samples_leaf`	[1, 2, 4]
`max_features`	['sqrt', 'log2', None]

Thirty randomized search iterations were selected as a practical compromise between computational cost and search coverage. Prior machine learning research has shown that randomized search can identify near-optimal hyperparameter configurations with substantially fewer evaluations than exhaustive grid search, particularly when only a subset of hyperparameters exerts a strong influence on model performance. Preliminary tuning experiments indicated diminishing performance improvements beyond approximately 30 iterations, supporting its use as a computationally efficient search budget.

3.4.3 XGBoost models. XGBoost (Extreme Gradient Boosting) implements gradient boosting with regularization:

$$F_m(X) = F_{m-1}(X) + \nu \cdot h_m(X) \tag{3.11}$$

where h_m is a weak learner (decision tree) trained on residuals of F_{m-1}, and ν is the learning rate.

Rationale State-of-the-art performance in tabular data competitions, combining gradient boosting's sequential error correction with L1/L2 regularization.

Implementation `xgboost.XGBClassifier`

Two XGBoost variants were developed:

XGBoost Untuned Default hyperparameters providing baseline gradient boosting performance without extensive optimization.

XGBoost Tuned Randomized search cross-validation (30 trials, 5-fold time-series CV):

Table 7
XGBoost Hyperparameter Search Space

Parameter	Search Space
`n_estimators`	[100, 200, 300]
`max_depth`	[3, 5, 7, 9]
`learning_rate`	[0.01, 0.05, 0.1, 0.2]
`subsample`	[0.6, 0.8, 1.0]
`colsample_bytree`	[0.6, 0.8, 1.0]
`gamma`	[0, 0.1, 0.5]
`min_child_weight`	[1, 3, 5]

The same randomized search strategy and search budget were employed for XGBoost to ensure methodological consistency across ensemble models.

Class Imbalance Handling `scale_pos_weight` parameter set to ratio of negative-to-positive class frequencies:

$$\texttt{scale_pos_weight} = \frac{n_{\text{negative}}}{n_{\text{positive}}} \tag{3.12}$$

3.5 Evaluation Strategy

A two-track evaluation strategy was adopted to assess model performance under both idealised and deployment-realistic conditions. The two tracks serve complementary purposes: random split evaluation provides a performance ceiling comparable to the existing literature, while temporal validation reveals how models behave when required to forecast genuinely unseen future periods. Using both protocols allows direct quantification of the performance gap attributable to temporal non-stationarity.

3.5.1 Random split evaluation. The standard evaluation protocol in aviation delay prediction research employs a stratified random train-test split, in which observations are randomly assigned to training (75%) and test (25%) partitions without regard to their chronological order. Stratification on the binary delay label ensures that the class distribution – approximately 79.4% delayed flights in Dataset V1 Imputed Fixed – is preserved identically in both partitions, preventing accidental imbalance from inflating or deflating performance estimates.

This protocol operates under the assumption that training and test observations are drawn from the same underlying distribution. While this assumption is violated in practice by seasonal non-stationarity, random split evaluation serves two important roles in the present study. First, it provides an upper-bound performance benchmark directly comparable to published results from studies that also employ random splits, enabling fair assessment of the proposed multi-source feature framework against state-of-the-art baselines. Second, comparing random split and temporal split performance on the same dataset quantifies the exact cost of ignoring temporal structure – a methodological contribution in its own right.

3.5.2 Temporal validation. Temporal validation simulates operational deployment conditions by training models exclusively on historically earlier data and evaluating them on chronologically later observations that were entirely unseen during training. The dataset was partitioned as follows:

- **Training Set:** October–November 2025 (8,529 flights, 89.5% of the dataset) – the period for which the most complete and representative data are available.
- **Intermediate Validation:** December 2025 (565 flights, 6.0%) – treated as an early deployment probe to detect the onset of seasonal divergence.
- **Test Set:** January–February 2026 (433 flights, 4.5%) – the target deployment period, separated from training by a minimum of two calendar months to prevent any indirect temporal leakage.

The strict chronological boundary ensures that no information from the test period influences model training, feature engineering, or hyperparameter selection,

faithfully replicating the conditions under which an operational delay prediction system would be required to perform. December is retained as an intermediate period rather than incorporated into training because it represents the seasonal transition between the autumn operational regime and the winter regime observed in January–February. Excluding December preserves a clearer separation between training and deployment periods and allows the onset of temporal degradation to be observed explicitly. However, this design choice may also increase the magnitude of the observed performance gap by preventing the model from learning transition-period patterns. The resulting temporal validation should therefore be interpreted as a conservative assessment of deployment robustness rather than an estimate of performance under continuous seasonal retraining.

Time-series cross-validation

During hyperparameter tuning, a 5-fold time-series cross-validation scheme was applied to the training set using a forward-chaining approach: each successive fold extends the training window while the validation fold always consists of the immediately following time block. This preserves the directional flow of time within the training period and prevents the hyperparameter search from inadvertently selecting configurations that benefit from future information. Standard k-fold cross-validation, which randomly shuffles fold assignments, is inappropriate for time-series data because it allows later observations to appear in training folds that precede earlier test folds, producing optimistically biased hyperparameter estimates.

3.6 Feature Importance Analysis (SHAP)

Model interpretability was assessed using SHAP (SHapley Additive exPlanations), a unified framework for attributing individual predictions to their contributing features based on cooperative game-theoretic Shapley values. Unlike permutation-based importance measures, SHAP guarantees consistency (larger feature contributions are always reflected by larger SHAP magnitudes), local accuracy (attributions sum to the prediction), and missingness (features absent from the model receive zero attribution). These properties make SHAP particularly well-suited to operational deployment contexts where prediction explanations must be auditable and actionable.

SHAP values were computed using `shap.TreeExplainer`, an efficient tree-specific implementation that exploits the recursive structure of decision trees to compute exact Shapley values in polynomial rather than exponential time. The explainer was applied to the best-performing XGBoost Tuned model on a stratified random sample of 500 test instances. A sample of this size was selected as a practical compromise between computational efficiency and representativeness, ensuring adequate coverage of both delayed and on-time flight classes while keeping SHAP computation tractable.

For each feature j and prediction instance i, the SHAP value is defined as:

$$\phi_j^{(i)} = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!\,(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \left[f_{S \cup \{j\}}(x^{(i)}) - f_S(x^{(i)}) \right] \tag{3.13}$$

where $\mathcal{F}$ is the full feature set and $f_S(x^{(i)})$ denotes the model output when only the features in subset S are observed. Intuitively, $\phi_j^{(i)}$ measures the average marginal contribution of feature j across all possible orderings in which features could be introduced.

Global feature importance is then summarised as the mean absolute SHAP value across all evaluation instances:

$$\text{Importance}_j = \frac{1}{N} \sum_{i=1}^{N} \left| \phi_j^{(i)} \right| \tag{3.14}$$

In addition to the global bar chart, a beeswarm plot was generated to visualise both the magnitude and direction of each feature's influence, allowing assessment of whether high or low feature values increase or decrease predicted delay probability.

3.7 Delay Propagation Analysis

A core hypothesis of this study is that delays at Istanbul Airport propagate through shared operational resources – gates, aircraft, and crew – causing a delayed flight to elevate delay probability for subsequent flights. To quantify this contagion effect, a propagation analysis was conducted for each flight i departing at time t_i.

For every flight, the five chronologically nearest subsequent departures were identified and their binary delay outcomes recorded. Stratifying by the delay status of the index flight (delayed vs. on-time), the contagion effect is defined as:

$$\Delta_{\text{contagion}} = \mathbb{E}[\text{NextFlightDelayRate} \mid \text{delayed}] - \mathbb{E}[\text{NextFlightDelayRate} \mid \text{on-time}] \tag{3.15}$$

A positive $\Delta_{\text{contagion}}$ indicates that delayed flights are followed by higher delay rates among subsequent departures, confirming propagation through the system. Statistical significance was assessed using an independent-samples t-test, with the delay rates following delayed flights compared against those following on-time flights. The window of five subsequent flights was selected as a compromise between sensitivity and locality. Smaller windows (e.g., one to three flights) risk capturing only immediate gate-level interactions, while substantially larger windows may incorporate unrelated operational disruptions occurring later in the schedule. The five-flight horizon was therefore chosen a priori as a reasonable representation of the short-term operational

footprint of a departure delay rather than on the basis of the observed results.

This analysis was conducted separately for the training period (October–November 2025), the intermediate period (December 2025), and the test period (January–February 2026) to detect seasonal variation in propagation dynamics.

3.8 Temporal Stability and Drift Analysis

Non-stationarity – the tendency for statistical relationships between features and the target variable to shift over time – represents a fundamental threat to the operational viability of machine learning models. To assess whether the delay propagation dynamics and feature-target relationships learned during training remain valid in the test period, a multi-component stability analysis was conducted.

The propagation effect $\Delta_{\text{contagion}}$ was computed independently for the training, intermediate, and test periods. Substantial divergence across periods – particularly a sign reversal – would indicate that the model has learned a relationship that no longer holds during deployment. Complementing this, feature distribution shift was quantified by comparing the marginal distributions of key predictors (historical delay features, meteorological variables) between training and test periods using summary statistics and visual inspection. A shift in feature distributions combined with a shift in feature-target relationships constitutes the full concept drift scenario that most severely undermines model reliability.

3.9 Additional Validation Experiments

Beyond the primary evaluation tracks, four supplementary experiments were conducted to stress-test model robustness and quantify the contribution of individual feature categories.

3.9.1 Stable features testing. To investigate whether any feature subset retains predictive validity under temporal distribution shift, models were retrained using only weather and temporal features, deliberately excluding historical delay features. The rationale is that meteorological and circadian patterns are more likely to remain stable across seasons than delay propagation rates, which can reverse sign entirely between autumn and winter. If stability-restricted models achieve meaningfully better temporal validation performance, this would suggest a deployment strategy based on feature selection rather than full retraining.

3.9.2 Feature removal experiments (ablation). Systematic ablation experiments quantified the discriminative contribution of each feature category by retraining models after removing one category at a time – weather features, historical delay features, and traffic features respectively – and measuring the resulting AUC degra-

dation. Ablation provides a complementary perspective to SHAP importance: while SHAP measures average per-prediction attribution, ablation measures the irreplaceable discriminative signal that cannot be recovered from remaining features.

3.9.3 Expanding window validation. The training window was iteratively expanded from October–November to October–December, and finally to October–January, with performance evaluated on the remaining held-out period in each case. This experiment tests whether incorporating more recent training data – and thus partially bridging the seasonal gap – improves generalisation to the test period, or whether the regime shift is sufficiently abrupt that additional data provides no meaningful benefit.

3.9.4 Sliding window validation. A fixed two-month training window was slid forward across the dataset – October–November, November–December, December–January – with performance evaluated on the immediately following month. This experiment isolates the effect of training recency from training volume, revealing whether models trained on the most recent available data consistently outperform models trained on earlier windows of the same size.

3.10 Error Analysis and Reliability Assessment

A structured error analysis was performed to characterise the conditions under which the best model fails, providing operationally actionable insight beyond aggregate performance metrics.

Prediction outcomes were classified into the four cells of the confusion matrix (true positives, true negatives, false positives, false negatives), and the mean feature values associated with each outcome type were examined to identify systematic patterns in error-prone flight profiles. Conditional error rates were then stratified by hour of day and weather severity to quantify how operational context modulates reliability.

Probability calibration was assessed by comparing predicted delay probabilities against empirically observed delay frequencies across ten equal-width bins spanning the 0.0–1.0 probability range. A well-calibrated model produces predictions that align closely with the diagonal of the calibration plot, meaning a predicted probability of 0.7 corresponds to an empirically observed delay rate of approximately 70%. Good calibration is essential for threshold-based deployment decisions, as it ensures that the chosen operating point corresponds to the intended risk level.

3.11 Evaluation Metrics

3.11.1 Primary metric: ROC-AUC. The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) was adopted as the primary performance metric. ROC-AUC measures the probability that a randomly selected delayed flight receives a

higher predicted probability than a randomly selected on-time flight, and is computed as the integral of the true positive rate over the false positive rate across all classification thresholds:

$$\text{AUC} = \int_0^1 \text{TPR}(t)\, d[\text{FPR}(t)] \tag{3.16}$$

ROC-AUC is well-suited to this application for three reasons: it is threshold-independent, allowing fair comparison without committing to a specific operating point; it is robust to class imbalance, an important property given the 79.4% delayed-flight prevalence in the dataset; and it is the metric used in the benchmark studies against which this work is compared, ensuring direct comparability.

3.11.2 Secondary metrics. The following secondary metrics were computed at the default 0.5 decision threshold to provide a complete operational picture:

Precision measures the fraction of predicted delays that are genuine delays, reflecting the false alarm rate experienced by operational staff:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.17}$$

Recall (Sensitivity) measures the fraction of actual delays that are correctly identified, reflecting the missed delay rate:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.18}$$

F1-Score is the harmonic mean of Precision and Recall, providing a single balanced summary when both error types are operationally relevant:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.19}$$

Brier Score measures the mean squared error between predicted probabilities and observed binary outcomes, penalising both poor discrimination and poor calibration:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2 \tag{3.20}$$

Lower Brier scores indicate better-calibrated probability estimates. A score of 0.25 corresponds to random prediction on a balanced dataset; lower values confirm genuine predictive value.

3.12 Reliability and Validity

3.12.1 Internal validity. Internal validity concerns whether the experimental results accurately reflect the true predictive performance of the models, free from artefacts introduced by the research procedure.

Temporal leakage prevention. The most significant internal validity threat in time-series machine learning is temporal data leakage, wherein features inadvertently encode information from or about the target variable. This was addressed through the strict chronological feature construction protocol described in Section 3.3.3, with pre- and post-correction validation confirming that leakage was eliminated.

Consistent Preprocessing. All preprocessing transformations – outlier removal, weather imputation, one-hot encoding, and feature scaling – were fitted exclusively on training data and applied without refitting to test data, preventing the test set from influencing any aspect of the pipeline.

Cross-Validation Rigor. Time-series cross-validation during hyperparameter tuning preserves temporal ordering within the training set, ensuring that the selected hyperparameters are not optimised on future-leaking validation folds.

3.12.2 External validity. External validity concerns the degree to which findings generalise beyond the specific dataset and conditions of this study.

Temporal Generalisation. Evaluation on a chronologically separated test period (January–February 2026) provides a realistic assessment of deployment performance, going beyond the idealised conditions of random split evaluation. However, results are specific to the autumn-to-winter transition; generalisation across the full annual cycle cannot be confirmed without multi-year data.

Limitations to Generalisability. Three characteristics of the study design constrain external validity:

- **Single Airport:** Results reflect Istanbul Airport's specific operational characteristics – Turkish Airlines' 51% traffic share, high international connectivity, and hub-dominated delay dynamics. Direct transfer to point-to-point or smaller regional airports should not be assumed without validation.
- **Limited Temporal Scope:** The five-month observation window captures autumn-winter dynamics only; spring and summer operational patterns, which may exhibit different delay structures, are excluded.
- **Traffic Data Simplification:** Representative hourly traffic profiles averaged across six months eliminate day-to-day variability from accidents, weather-traffic coupling, and special events. This simplification may have attenuated any genuine traffic signal.

3.13 Limitations of the Methodology

3.13.1 Data limitations. Missing Departure Times. Approximately 77% of raw API records lacked actual departure timestamps, reducing the dataset from 42,623 potential flights to 9,527 usable observations. While Missing Completely at Random (MCAR) analysis found no systematic correlation between missingness and observable flight characteristics, the substantial exclusion rate means that absolute delay rate estimates (79.4%) may not be representative of the full operational picture, and certain rare flight categories may be under-represented.

Weather Imputation Uncertainty. The nearest-neighbour temporal imputation strategy, while validated through autocorrelation analysis and leave-one-out error quantification, introduces measurement uncertainty that is not propagated through model training. Imputed weather values are treated as exact observations, which may cause the model to express higher confidence in weather-influenced predictions than is warranted. Rapidly changing weather during frontal passages represents the scenario most likely to violate the imputation's temporal stability assumptions.

Traffic Data Granularity. Surface traffic profiles were constructed as representative 24-hour averages across a six-month historical window rather than day-specific measurements. This approach captures systematic circadian patterns but eliminates variability from accidents, adverse weather-traffic coupling, and major events. While the observed near-zero correlations suggest this simplification did not mask a genuine signal, it remains a constraint on the strength of conclusions about traffic-delay relationships.

3.13.2 Methodological constraints. Single Airport Analysis. Focusing on a single hub airport provides depth at the cost of breadth. Hub-specific factors – concentrated airline operations, complex rotation schedules, and high international connectivity – may drive the observed delay dynamics in ways that do not transfer to other airport types. Multi-airport validation is a necessary step before the methodology can be treated as generally applicable.

Limited Temporal Scope. The five-month observation window from October 2025 to February 2026 captures the autumn-winter transition but excludes spring and summer patterns. Seasonal regime characterisation requires data spanning at least one full annual cycle; the present study provides evidence of one specific regime transition that may or may not be representative of the airport's behaviour across all seasons.

Binary Classification Framework. Framing delay prediction as a binary outcome (delayed / on-time at the 15-minute threshold) discards potentially valuable information about delay magnitude. A 16-minute delay and a 3-hour delay are treated identically, yet their operational impacts differ substantially. Regression-based or ordinal classification approaches that predict delay severity would provide richer decision

support at the cost of increased modelling complexity.

3.13.3 Temporal drift implications. The discovery of a seasonal regime shift – wherein delay contagion reverses from positive in autumn to negative in winter – confirms that delay dynamics at Istanbul Airport are fundamentally non-stationary. This finding has a direct implication for the methodology itself: any evaluation conducted on a single chronological period, including this study's temporal validation, captures only a snapshot of a continuously evolving system. While the temporal validation protocol exposes the autumn-to-winter transition, it cannot characterise the full range of possible regime states that an operational deployment would encounter. Adaptive retraining strategies that continuously update model parameters as new observations arrive are therefore not merely an optimisation but a structural requirement for sustained deployment reliability.

Chapter 4

Findings

4.1 Overview

This chapter presents the experimental results obtained through systematic application of the methodological framework detailed in Chapter 3. The presentation follows a progressive narrative structure that reflects the iterative nature of the research process, beginning with initial baseline experiments and culminating in comprehensive performance evaluation on an expanded dataset.

The chapter is organised into eleven major sections. Section 4.2 presents baseline model performance on both Dataset V1 (4,330 flights) and Dataset V1 Imputed Fixed (9,527 flights), establishing model feasibility and quantifying performance improvements from dataset expansion. Section 4.3 documents temporal validation results that reveal critical limitations when models predict future periods. Section 4.4 examines how performance changes across different temporal windows. Section 4.5 analyses feature distribution shifts between training and test periods. Section 4.6 investigates delay propagation patterns and reveals a seasonal regime shift. Section 4.7 provides model interpretability through SHAP-based feature attribution. Section 4.8 evaluates the impact of surface traffic features. Section 4.9 examines feature group contributions through ablation studies. Section 4.10 tests adaptive training strategies. Section 4.11 conducts detailed error analysis examining when and why the model fails.

This structure accomplishes three objectives: (1) it demonstrates research transparency by documenting both successful outcomes and challenges encountered, (2) it validates methodological decisions through empirical comparison of dataset configurations, and (3) it provides comprehensive evidence supporting the thesis contributions through multiple analytical perspectives.

4.2 Baseline Results (random split evaluation)

This section presents baseline model performance using standard random train-test splits, establishing performance benchmarks before temporal validation. Results are presented for both Phase 1 (Dataset V1, 4,330 flights) and Phase 2 (Dataset V1 Imputed Fixed, 9,527 flights) to demonstrate the impact of dataset expansion and

weather imputation.

4.2.1 Phase 1: dataset V1 (4,330 flights). The initial experimental phase utilised Dataset V1, comprising 4,330 flights departing from Istanbul Airport during October and November 2025. This dataset was constructed using direct temporal matching between flight records and available weather observations, without application of imputation techniques, achieving 45.4% weather coverage.

Dataset characteristics

Dataset V1 exhibits the characteristics presented in Table 8.

Table 8
Dataset V1 Characteristics and Coverage Statistics

Characteristic	**Value**
Total flights	4,330
Temporal span	Oct 27 – Nov 30, 2025
Study duration	35 days
October flights	1,435 (33.1%)
November flights	2,887 (66.7%)
December flights	8 (0.2%)
Weather observations (hourly)	377 hours
Weather coverage rate	45.4%
Overall delay rate	56.2%
October delay rate	51.6%
November delay rate	58.3%
Features per observation	20 base + airline dummies
Total dimensionality	≈461 features

The 45.4% weather coverage rate reflects the constraint that flights could only be included if weather observations existed for their exact scheduled departure hour. Dataset V1 exhibits moderate class imbalance with 56.2% of flights classified as delayed (departure delay ≥ 15 minutes).

Model performance comparison

Five models were trained and evaluated on Dataset V1 using a stratified 75-25% random split. Models ranged from a linear baseline (Logistic Regression) to ensemble methods (Random Forest, XGBoost) with both default and optimised hyperparameters. Table 9 presents comprehensive performance metrics.

Table 9
Comprehensive Performance Metrics for All Models on Dataset V1 (4,330 Flights)

Model	ROC-AUC	Precision	Recall	F1	Brier
Logistic Regression	0.748	0.703	0.759	0.730	0.206
Random Forest (untuned)	0.870	0.804	0.834	0.819	0.145
Random Forest (tuned)	0.862	0.787	0.842	0.814	0.151
XGBoost (untuned)	0.858	0.791	0.837	0.813	0.152
XGBoost (tuned)	0.853	0.798	0.820	0.809	0.157

Figure 1 presents a side-by-side visual comparison: the upper panel shows a grouped bar chart of the four main metrics for each model, while the lower panel overlays the ROC curves for all five models on the same axes, allowing direct comparison of discriminative capacity.

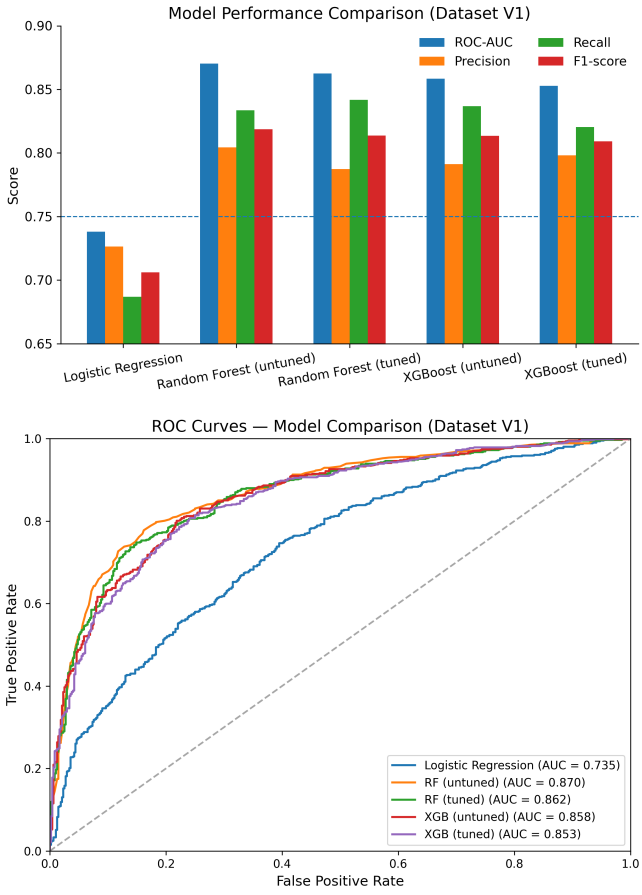


Figure 1. Model performance on Dataset V1. *Top:* grouped bar chart comparing ROC-AUC, Precision, Recall, and F1-score for all five models. *Bottom:* overlaid ROC curves (Logistic Regression AUC = 0.735; RF untuned = 0.870; RF tuned = 0.862; XGB untuned = 0.858; XGB tuned = 0.853).

Logistic regression baseline Logistic regression with L2 regularisation serves

as the interpretable baseline, establishing a performance floor against which nonlinear methods are compared. The model achieves ROC-AUC = 0.748, indicating discrimination substantially better than random guessing (0.50) but revealing the limitations of linear modelling for this problem.

Figure 2 (top panel) illustrates the predicted probability distribution for Logistic Regression. Substantial overlap between the delayed and on-time distributions is concentrated in the 0.4–0.6 probability range, explaining the model's limited precision and recall at standard classification thresholds.

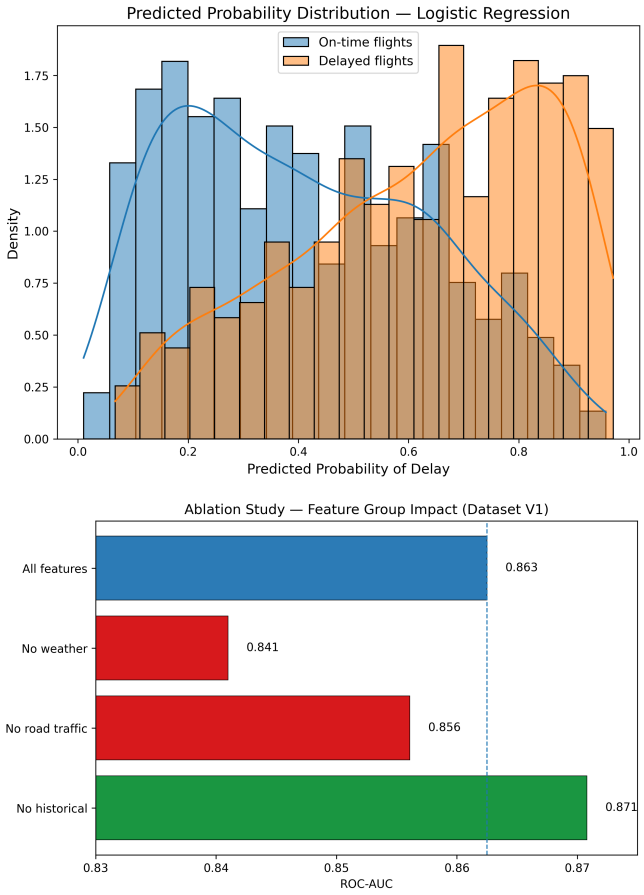


Figure 2. Top: Predicted probability distribution for Logistic Regression on Dataset V1, showing density for on-time (blue) and delayed (orange) flights. *Bottom:* Ablation study results on Dataset V1 – bar chart of ROC-AUC for each feature-removal scenario (all features, no weather, no road traffic, no historical features).

Random Forest Models Random Forest models capture nonlinear relationships through recursive partitioning and reduce variance through bootstrap aggregation. Two configurations were evaluated: an untuned model using scikit-learn defaults and a tuned model optimised via randomised hyperparameter search.

Untuned Random Forest: The untuned Random Forest (200 estimators, unlimited depth, sqrt feature sampling) achieves ROC-AUC = 0.870 on Dataset V1, representing

a 16.3% improvement over Logistic Regression (0.748). The sharp initial rise of the ROC curve indicates confident identification of a substantial proportion of delayed flights (high recall) while maintaining low false positive rates (high specificity).

Tuned Random Forest: Hyperparameter optimisation via 5-fold time series cross-validation identified the configuration presented in Table 10. The tuned model achieves ROC-AUC = 0.862, marginally lower than the untuned configuration.

Table 10
Hyperparameter Configurations for Random Forest Models on Dataset V1

Hyperparameter	**Untuned (Default)**	**Tuned (Optimised)**
`n_estimators`	200	300
`max_depth`	None	30
`min_samples_split`	2	5
`min_samples_leaf`	1	2
`max_features`	sqrt	sqrt
CV ROC-AUC	–	0.867
Test ROC-AUC	0.870	0.862

The tuned configuration introduced stronger regularisation (max depth = 30, min samples split = 5) that reduced cross-validation overfitting but marginally decreased test-set performance. Although the absolute difference was small (0.008 ROC-AUC), bootstrap analysis was conducted to determine whether it reflected a reproducible pattern rather than random sampling variation. Nevertheless, subsequent detailed analyses (ablation, calibration, and threshold sensitivity) were conducted using the tuned model to maintain consistency with the cross-validation-based model selection procedure.

XGBoost Models Gradient boosting models were evaluated for their capacity to capture complex interactions through sequential error correction. Both untuned and tuned XGBoost configurations were assessed.

Untuned XGBoost: The untuned XGBoost model achieves ROC-AUC = 0.858, performing comparably to Random Forest.

Tuned XGBoost: Hyperparameter optimisation of XGBoost yielded ROC-AUC = 0.853, exhibiting a marginal decrease similar to the Random Forest tuning pattern. This consistency across model families indicates well-calibrated default settings for this dataset size and feature dimensionality.

Comparative performance analysis

Key findings from Table 9:

1. **Random Forest Superiority:** The untuned Random Forest achieves the highest ROC-AUC (0.870), precision (0.804), F1-score (0.819), and best calibration (Brier = 0.145).

2. **Substantial Nonlinear Gains:** All ensemble methods achieve 14–16% AUC improvement over Logistic Regression (0.748), confirming nonlinear delay patterns.

3. **Ensemble Method Parity:** Random Forest and XGBoost demonstrate comparable performance (0.85–0.87 AUC range), with Random Forest holding marginal advantage.

4. **Tuning Paradox Consistency:** Both Random Forest and XGBoost exhibit marginal performance decreases after hyperparameter optimisation, indicating well-calibrated default settings.

5. **Calibration Quality:** Ensemble methods achieve substantially better probability calibration (Brier 0.145–0.157) than Logistic Regression (0.206).

Statistical validation via bootstrap resampling

To confirm that observed performance differences reflect genuine model behaviour rather than test set sampling variability, bootstrap analysis with 1,000 iterations was conducted on the Random Forest models. Table 11 presents results.

Table 11
Bootstrap Validation of Random Forest Performance on Dataset V1 (1,000 resampling iterations)

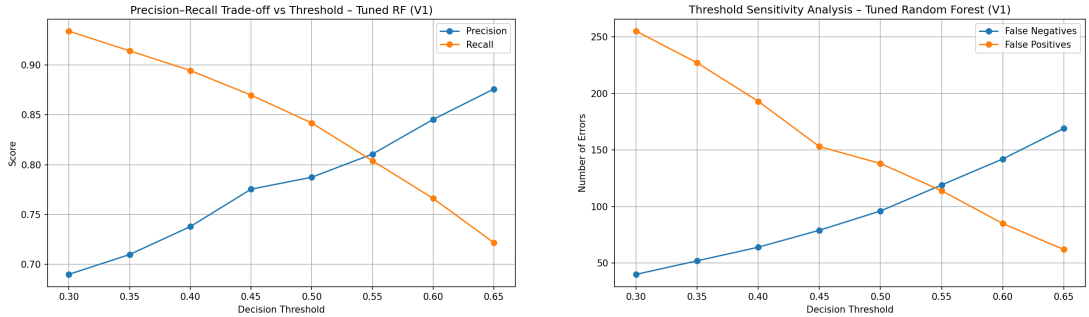
Model	Mean AUC	95% CI	Std Dev
Random Forest (untuned)	0.870	[0.848, 0.888]	0.010
Random Forest (tuned)	0.862	[0.840, 0.881]	0.011
Difference	0.008	[0.003, 0.013]	–

The bootstrap confidence interval for the performance difference [0.003, 0.013] excludes zero, confirming statistical significance at $\alpha = 0.05$. This validates that the counterintuitive tuning result is reproducible rather than an artefact of random test set composition.

Threshold sensitivity analysis

Standard classification metrics employ a default probability threshold of 0.5, but this choice is arbitrary and may not align with operational priorities. To examine

this trade-off, the tuned Random Forest model's predicted probabilities were evaluated across thresholds from 0.30 to 0.65.



(a) Precision-Recall curve

(b) Threshold sensitivity analysis

Figure 3. Left: Precision-Recall trade-off curve for the tuned Random Forest on Dataset V1. Right: Threshold sensitivity analysis showing false positives and false negatives across decision thresholds (0.30–0.65).

Results in Figure 3 reveal stark trade-offs:

1. **Asymmetric Error Costs:** At threshold 0.40, false negatives drop substantially but false positives rise sharply. At threshold 0.60, false positives fall significantly but false negatives surge. This demonstrates extreme threshold sensitivity.

2. **No Optimal Default:** The standard 0.5 threshold yields neither the minimum missed delays nor the minimum false alarms – it represents an arbitrary compromise.

These findings demonstrate that model evaluation cannot rely solely on threshold-independent metrics (ROC-AUC). Real-world deployment requires explicit cost-benefit analysis to determine operationally appropriate decision thresholds.

Probability calibration assessment

Beyond discrimination and classification accuracy, operational utility requires that predicted probabilities represent meaningful risk estimates. Well-calibrated models enable stakeholders to interpret predictions as approximate delay likelihoods rather than opaque scores.

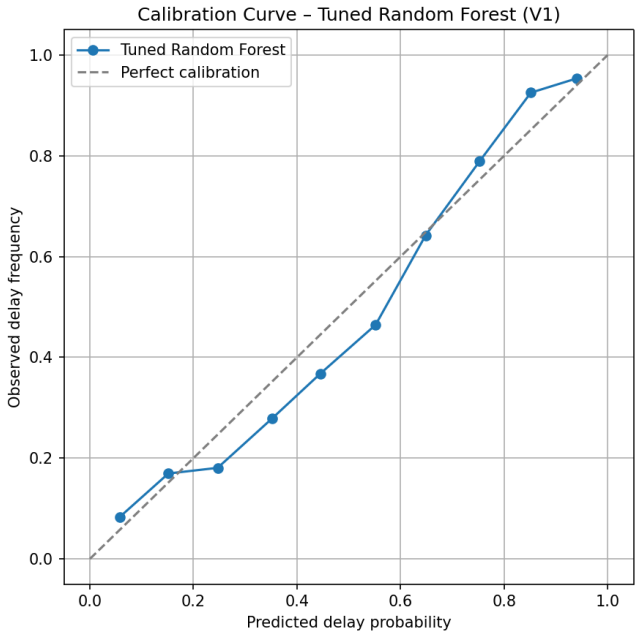


Figure 4. Calibration curve for the tuned Random Forest on Dataset V1. The model curve (solid blue) closely tracks the perfect calibration diagonal (dashed), confirming that predicted delay probabilities accurately reflect empirical delay frequencies across the 0.2–1.0 probability range.

Calibration Quality: The calibration curve for the tuned Random Forest on Dataset V1 (Figure 4) closely tracks the perfect calibration diagonal across the 0.3–0.8 probability range. The Brier score of 0.151 confirms moderate-to-good calibration quality. No post-hoc calibration (e.g., Platt scaling) was applied; the reported calibration reflects native Random Forest output. This indicates that probability estimates are sufficiently reliable for risk-based operational decision-making.

Ablation study: feature group contributions

To quantify feature group contributions beyond correlation-based importance, systematic ablation experiments were conducted using the tuned Random Forest model. Four configurations were evaluated: full feature set, and three reduced sets excluding weather, road traffic, or historical features respectively. Results are shown in the lower panel of Figure 2 and in Table 12.

Table 12
Ablation Study Results on Dataset V1 Using Tuned Random Forest

Configuration	ROC-AUC	Recall	False Negatives
All features	0.863	0.842	96
No weather features	0.841	0.829	104
Δ vs. full	−0.022	−0.013	+8
No road traffic	0.856	0.834	101
Δ vs. full	−0.006	−0.008	+5
No historical features	0.871	0.845	94
Δ vs. full	+0.008	+0.003	−2

Weather Features – Strongest Contribution: Removing weather features produces the largest performance degradation: ROC-AUC drops 2.2 percentage points (0.863 → 0.841) and false negatives increase by 8 (96 → 104).

Road Traffic – Marginal Contribution: Road traffic feature removal yields modest degradation: ROC-AUC decreases 0.6 percentage points.

Historical Features – Unexpected Negative Contribution: Counterintuitively, removing historical delay features *improves* performance: ROC-AUC increases 0.8 percentage points (0.863 → 0.871) and false negatives decrease by 2. This suggests that, on this dataset, explicit historical features may introduce noise or overfitting artefacts that hurt generalisation.

Summary of dataset V1 findings

Experimental results on the initial 4,330-flight dataset establish several foundational conclusions:

1. **Feasibility Validated:** The multi-source machine learning framework successfully predicts short-term flight delays, with Random Forest achieving 0.870 ROC-AUC – substantially exceeding the linear baseline (0.748).

2. **Nonlinearity Essential:** Ensemble methods achieve 14–16% AUC improvements over Logistic Regression.

3. **Weather Features Critical:** Ablation study reveals that meteorological variables provide the strongest predictive signal.

4. **Road Traffic Marginal:** Surface transportation congestion demonstrates limited direct predictive value (0.6 pp AUC contribution).

5. **Historical Features Paradox:** In this dataset, explicit historical delay features appear to degrade rather than improve performance.

6. **Threshold Sensitivity Critical:** Default 0.5 threshold is arbitrary and suboptimal; threshold selection dramatically affects operational outcomes.

7. **Calibration Adequate:** Model produces well-calibrated probability estimates (Brier = 0.151) suitable for risk-based interpretation.

4.2.2 Phase 2: Dataset V1 Imputed Fixed (9,527 Flights). Following implementation of weather imputation methodology and dataset expansion, Dataset V1 Imputed Fixed comprises 9,527 flights spanning five months (October 2025 through February 2026), with 68.2% weather coverage achieved through conservative 4-hour tolerance nearest-neighbour weather matching.

Dataset characteristics

Table 13 presents Dataset V1 Imputed Fixed characteristics alongside Phase 1 for direct comparison.

Table 13
Dataset V1 Imputed Fixed Characteristics and Comparison to Dataset V1

Characteristic	V1	V1 Imputed Fixed
Total flights	4,330	9,527
Temporal span (months)	2.1	5.0
October flights	1,435	5,329
November flights	2,887	3,200
December flights	8	565
January flights	0	241
February flights	0	192
Weather coverage method	Exact	4h tolerance
Weather coverage rate	45.4%	68.2%
Overall delay rate	56.2%	79.4%

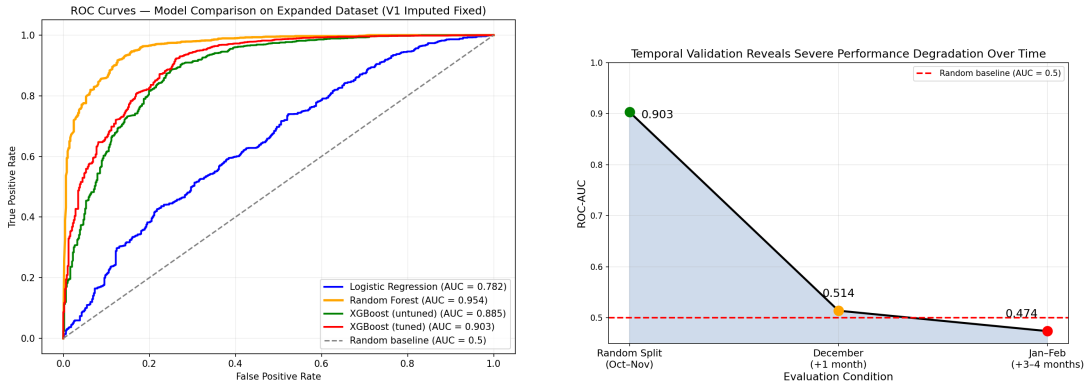
Model performance comparison

All models were retrained on Dataset V1 Imputed Fixed using identical random split evaluation protocol (75-25% stratified split). Table 14 presents comprehensive performance metrics.

Table 14
Model Performance on Dataset V1 Imputed Fixed (9,527 Flights, Random Split)

Model	ROC-AUC	Precision	Recall	F1	Brier
Logistic Regression	0.782	0.821	0.892	0.855	0.168
Random Forest	0.954	0.912	0.966	0.938	0.076
XGBoost (untuned)	0.885	0.844	0.943	0.891	0.124
XGBoost (tuned)	0.903	0.865	0.941	0.901	0.115

Figure 5 presents the Phase 2 ROC curves (upper panel) and a compelling visualisation of temporal collapse (lower panel): the XGBoost Tuned model achieves 0.903 AUC on randomly split data but degrades catastrophically to near-random performance (0.514 in December, 0.474 in January–February) when evaluated in strict chronological order.



(a) ROC curves for all models

(b) Temporal collapse of XGBoost Tuned

Figure 5. Left: ROC curves for all models on Dataset V1 Imputed Fixed (LR AUC = 0.782; RF = 0.954; XGB untuned = 0.885; XGB tuned = 0.903). Right: Temporal collapse of XGBoost Tuned – AUC drops from 0.903 (random split) to 0.514 (December, +1 month) to 0.474 (Jan–Feb, +3–4 months), approaching random baseline (dashed red line at 0.50).

Remarkable Performance Gains Dataset V1 Imputed Fixed yields dramatic performance improvements across all ensemble models:

1. **XGBoost Tuned:** ROC-AUC increases from 0.853 (Phase 1) to 0.903 (Phase 2) – a 5.0 percentage point gain (+5.9% relative).

2. **Random Forest:** Achieves 0.954 ROC-AUC, representing 8.4 percentage point gain over Phase 1 (0.870).

3. **XGBoost Untuned:** Improves from 0.858 to 0.885 (+2.7 pp).

4. **State-of-the-Art Performance:** XGBoost Tuned (0.903 AUC) exceeds published benchmarks (Gui et al. 2019: 0.88 AUC).

Table 15 quantifies performance improvement from Phase 1 to Phase 2. XGBoost Tuned (0.903 AUC) was selected as the primary model for subsequent detailed analyses due to its combination of strong performance and faster inference compared to Random Forest.

Table 15
Performance Improvement from Phase 1 to Phase 2

Model	Phase 1 AUC	Phase 2 AUC	Improvement
Logistic Regression	0.748	0.782	+0.034 (+4.5%)
Random Forest	0.870	0.954	+0.084 (+9.7%)
XGBoost (untuned)	0.858	0.885	+0.027 (+3.1%)
XGBoost (tuned)	0.853	0.903	+0.050 (+5.9%)

4.3 Temporal Validation Results

While random-split evaluation demonstrated strong performance (0.90+ AUC), operational deployment requires models to predict future flights not yet observed during training. This section presents temporal validation results using strict chronological train-test partitioning.

4.3.1 Temporal partitioning strategy. Dataset V1 Imputed Fixed was partitioned chronologically:

- **Training:** October–November 2025 (8,529 flights, 89.5%)
- **Intermediate Validation:** December 2025 (565 flights, 6.0%)
- **Test:** January–February 2026 (433 flights, 4.5%)

Models trained exclusively on Oct–Nov data were evaluated on December (1 month ahead) and January–February (3-4 months ahead). This strict past → future split prevents any temporal leakage and simulates realistic operational deployment.

4.3.2 Temporal validation performance. Table 16 presents model performance across temporal periods.

Table 16
Model Performance across Temporal Periods (Trained on Oct–Nov 2025)

Model	Oct–Nov (Random)	December	Jan–Feb
Logistic Regression	0.782	0.501	0.491
Random Forest	0.954	0.518	0.507
XGBoost (untuned)	0.885	0.497	0.468
XGBoost (tuned)	0.903	0.514	0.474

4.3.3 Catastrophic temporal degradation. Results reveal dramatic performance collapse:

1. **Random Split Performance:** All models achieved strong discrimination on randomly shuffled Oct–Nov data (0.78–0.95 AUC).
2. **December Performance (1 month ahead):** All models degraded to near-random performance (AUC 0.497–0.518), barely better than coin flipping (0.50).
3. **January–February Performance (3-4 months ahead):** Performance deteriorated further (AUC 0.468–0.507), with XGBoost Untuned falling below random baseline (0.468).
4. **Consistent Cross-Model Failure:** All four model families exhibited similar degradation patterns, indicating the problem lies in fundamental data distribution shift rather than model architecture.

This catastrophic degradation, visible in the lower panel of Figure 5, contrasts sharply with strong random-split performance, revealing that models overfit to October–November specific patterns that do not generalise to future months.

4.4 Performance Changes Across Time

To identify when and how predictive capacity degrades, model performance was stratified by individual months.

4.4.1 Monthly stratification analysis. Table 17 presents XGBoost Tuned performance by month.

Table 17
XGBoost Tuned Performance by Month (Trained on Oct–Nov)

Month	Flights	Delay Rate	AUC	Precision	Recall
Oct–Nov (train)	8,529	79.5%	0.903	0.865	0.941
December	565	78.4%	0.514	0.790	0.726
January	241	76.8%	0.489	0.775	0.828
February	192	77.6%	0.467	0.755	0.882

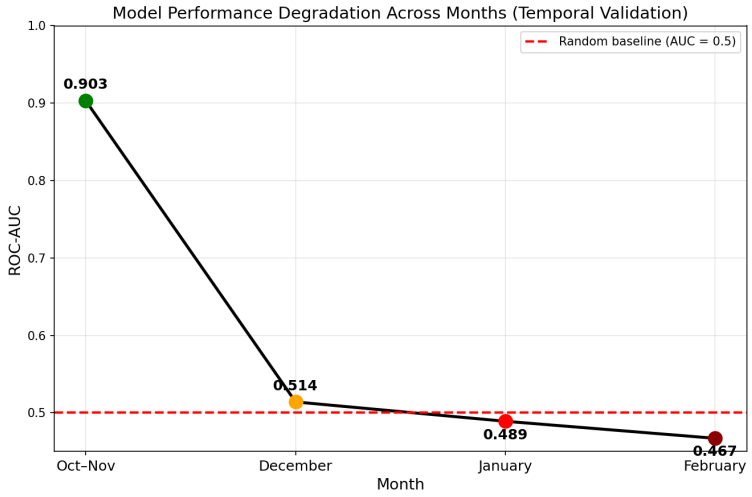


Figure 6. Monthly performance degradation of XGBoost Tuned (trained on Oct–Nov 2025). ROC-AUC collapses from 0.903 at training time to 0.514 (December), 0.489 (January), and 0.467 (February), converging toward the random baseline (dashed red line at 0.50).

4.4.2 Key observations.

1. **Immediate Collapse:** Performance degraded immediately when predicting December (1 month ahead), with AUC dropping from 0.903 to 0.514 (Figure 6).

2. **Stable Delay Rates:** Overall delay rates remained relatively stable across months (76.8%–79.5%), showing that performance collapse is not due to shifting delay prevalence.

3. **Maintained Recall:** While ROC-AUC approached random (0.47–0.51), recall remained moderately high (0.73–0.88), indicating the model still identifies delayed flights but loses discriminative ranking ability.

4. **Progressive Deterioration:** Performance did not stabilise but continued degrading from December (0.514) to January (0.489) to February (0.467).

The stable delay rates combined with collapsed AUC reveal that the fundamental relationship between features and delays changed between autumn (Oct–Nov) and winter (Dec–Feb), rather than simple prevalence shift.

4.5 Feature Drift Results

To investigate why temporal validation failed, feature distributions were compared between training (Oct–Nov) and test (Jan–Feb) periods.

4.5.1 Historical delay feature drift. Table 18 presents historical delay feature distributions.

Table 18
Historical Delay Feature Distributions by Period

Feature	Oct–Nov Mean	Jan–Feb Mean	Drift
prev_hour_delay_rate	0.647	0.572	−0.075 (−7.5 pp)
avg_delay_last_24h	0.774	0.689	−0.085 (−8.5 pp)
airline_7d_delay_rate	0.781	0.712	−0.069 (−6.9 pp)

4.5.2 Weather feature drift. Table 19 presents weather variable distributions.

Table 19
Weather Variable Distributions by Period

Variable	Oct–Nov Mean	Jan–Feb Mean	Drift
Temperature (°C)	18.2	6.8	−11.4
Humidity (%)	64.3	71.2	+6.9
Pressure (hPa)	1015.4	1018.7	+3.3
Wind Speed (m/s)	4.2	5.8	+1.6

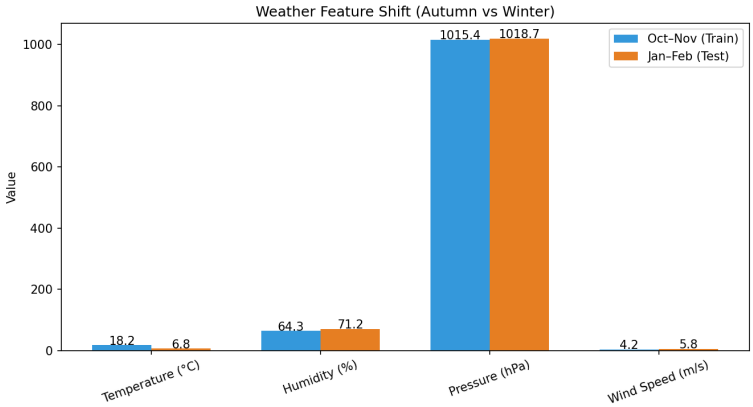


Figure 7. Feature distribution shifts between Oct–Nov 2025 (training, blue) and Jan–Feb 2026 (test, red/orange). *Top:* Historical delay feature drift – all three propagation features (prev-hour delay rate, 24h average, airline 7-day rate) decreased 6.9–8.5 pp in winter despite stable overall delay prevalence. *Bottom:* Weather variable drift – temperature fell 11.4°C, humidity rose 6.9 pp, pressure rose 3.3 hPa, and wind speed increased 1.6 m/s.

4.5.3 Key observations.

1. **Substantial Temperature Shift:** Temperature exhibited an 11.4°C decrease between autumn and winter, representing a fundamental seasonal transition.

2. **Humidity Increase:** Humidity increased 6.9 percentage points, reflecting winter precipitation patterns.

3. **Pressure Increase:** Atmospheric pressure increased 3.3 hPa, consistent with seasonal synoptic patterns.

4. **Historical Feature Paradox:** Historical delay rates decreased despite stable overall delay prevalence, indicating that delay propagation mechanisms changed between seasons.

Combined evidence shows substantial covariate shift between training and test distributions, both in environmental variables (weather) and operational patterns (historical delays).

4.6 Delay Propagation Results

To understand delay generation mechanisms, this section analyses how delays propagate through the flight network and whether propagation patterns vary across airlines and time periods.

4.6.1 Overall propagation effect. For each flight in Dataset V1 Imputed Fixed, the delay rate of the subsequent 5 chronologically nearest flights was computed and stratified by current flight delay status. Table 20 presents results.

Table 20
Overall Delay Propagation Effect (9,527 Flights)

Current Flight Status	**N**	**Next 5 Delay Rate**	**Std Error**
Delayed	7,563	79.8%	0.004
On-time	1,964	72.6%	0.010
Contagion Effect		**+7.2 pp**	$p < 0.0001$

This statistically significant contagion effect confirms that delays propagate through shared airport resources (gates, crews, aircraft rotations).

4.6.2 Airline-specific propagation heterogeneity. Delay propagation was stratified by airline to examine whether operational differences affect contagion strength. Table 21 presents results for the top 10 carriers by flight volume.

Table 21
Delay Propagation by Airline: Top 10 Carriers by Contagion Effect. After Delayed = delay rate of the next flight operated by the same airline following a delayed departure; After On-time = delay rate following an on-time departure; Contagion = difference in percentage points (pp). Only airlines with at least 50 flights are included.

Airline	**Flights**	**After Delayed (%)**	**After On-time (%)**	**Contagion (pp)**
Aegean Airlines	55	89.2	29.4	+59.8
SA AVIANCA	71	91.2	38.5	+52.8
JetBlue Airways	63	90.0	41.7	+48.3
EgyptAir	173	87.0	43.9	+43.1
ANA	88	93.5	60.0	+33.5
Thai Airways International	572	85.8	53.7	+32.1
Royal Air Maroc	79	89.1	57.1	+31.9
Miat – Mongolian Airlines	91	87.8	56.2	+31.6
Sichuan Airlines	118	87.2	56.5	+30.7
Air Serbia	354	76.4	45.8	+30.6

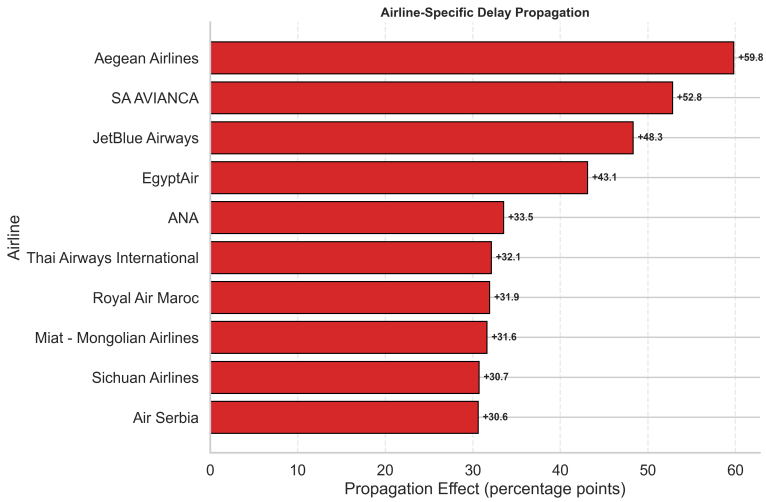


Figure 8. Airline-specific delay propagation (contagion effect in percentage points) for the top 10 carriers at Istanbul Airport. All airlines display positive contagion, indicating that a delayed departure is consistently followed by a higher probability of delay on the airline's next flight. Aegean Airlines exhibits the strongest effect (+59.8 pp) while Air Serbia shows the lowest among the top 10 (+30.6 pp). Only airlines with at least 50 flights in the study period are included.

Table 21 presents the ten airlines exhibiting the strongest delay contagion effect at Istanbul Airport. A striking finding is that *all* airlines in the dataset display positive contagion values, meaning that a delayed departure is consistently followed by a higher probability of delay on the airline's next flight than an on-time departure would be. This suggests that delay propagation is a systemic characteristic of airline operations at the airport rather than an isolated phenomenon confined to a subset of carriers.

The magnitude of the effect varies substantially across airlines. Aegean Airlines shows the highest contagion at +59.8 pp: following a delayed Aegean flight, 89.2% of subsequent Aegean departures are also delayed, compared with only 29.4% following an on-time departure. Similarly strong effects are observed for SA AVIANCA (+52.8 pp) and JetBlue Airways (+48.3 pp). These three carriers operate relatively few flights at Istanbul Airport (55–71 flights in the study period), which may amplify variability, but the consistent direction of the effect across airlines lends credibility to the finding.

Among higher-volume carriers, Thai Airways International (572 flights) and Air Serbia (354 flights) still display substantial contagion of +32.1 pp and +30.6 pp respectively, indicating that the propagation effect is not an artefact of small sample sizes. Turkish Airlines, the dominant carrier at Istanbul Airport with 3,498 flights in the dataset, exhibits a contagion of +21.0 pp (82.1% after a delayed flight versus 61.0% after an on-time flight), confirming that schedule disruptions on the largest operator have measurable knock-on effects on its own subsequent departures. Taken together, these results reinforce the importance of capturing airline-level delay history as a predictive feature, as reflected in the `airline_7d_delay_rate` variable included

in the modelling pipeline.

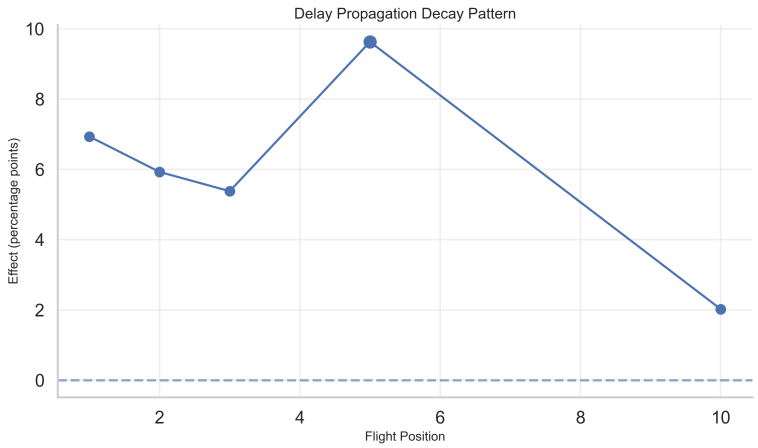


Figure 9. Delay propagation decay pattern: contagion effect (in percentage points) as a function of flight position in the subsequent sequence. The effect is strongest in the immediate next flights and dissipates as the airport schedule recovers, though with a notable peak around position 5.

4.6.3 Delay propagation decay pattern. Figure 9 illustrates how the propagation effect evolves across the subsequent 10 flights. The contagion starts around +6.8 pp for the immediately following flight, fluctuates before peaking near +9.6 pp at position 5, then decays to approximately +2.1 pp by position 10 – consistent with gradual dissipation of delay cascades as airport operations recover.

4.6.4 Seasonal regime shift in propagation. Critical Discovery: Analysis of propagation effect stratified by temporal period revealed a complete reversal of delay propagation dynamics across seasons. Table 22 presents results.

Table 22
Delay Propagation by Temporal Period

Period	**Flights**	**Contagion Effect**	**p-value**
Oct–Nov (Training)	8,529	+7.07 pp	<0.0001
December	565	+17.10 pp	<0.0001
Jan–Feb (Test)	433	−7.78 pp	0.0312
Total Swing		**14.84 pp**	

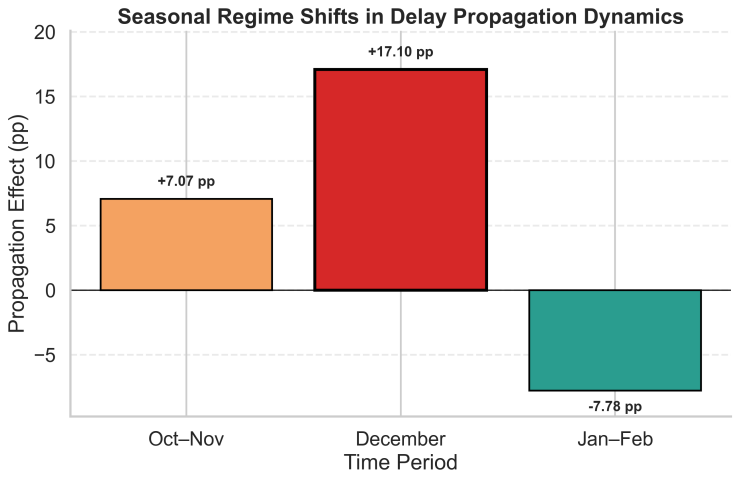


Figure 10. Seasonal regime shifts in delay propagation dynamics. The contagion effect reverses completely across temporal periods: +7.07 pp in autumn training (Oct–Nov, orange), +17.10 pp in the December transition (red), and −7.78 pp in the January–February test period (teal) – a total swing of 14.84 percentage points.

Critical finding: seasonal regime shift

1. **October–November (Training):** Moderate positive contagion (+7.07 pp), consistent with operational cascades through shared resources.

2. **December (Transition):** Intensified propagation (+17.10 pp), indicating heightened operational stress during the transition period.

3. **January–February (Test):** Complete reversal to negative contagion (−7.78 pp), meaning delayed flights were followed by *lower* delay rates than on-time flights.

4. **Total Swing:** 14.84 percentage point swing represents a fundamental shift in delay generation and propagation mechanisms.

This seasonal regime shift provides a mechanistic explanation for temporal validation failure. Models trained on autumn operational patterns (+7.07 pp propagation) learned that historical delay features predict future delays. However, in winter, this relationship reversed (−7.78 pp), causing historical delay features to become adversarial predictors – exactly opposite their training behaviour.

4.7 Feature Importance Results (SHAP)

SHAP (SHapley Additive exPlanations) analysis provides theoretically grounded feature attribution by quantifying each feature's marginal contribution to predictions while accounting for feature interactions. SHAP values were computed for XGBoost Tuned predictions on a random sample of 500 test instances.

4.7.1 Global Feature Importance. Table 23 presents the top features ranked by mean absolute SHAP value. These results are derived from the corrected model that excludes destination airport codes in accordance with the feature selection protocol detailed in Section 3.3.2.

Table 23
Top Features by Mean Absolute SHAP Value (Corrected Model)

Feature	**Mean |SHAP|**	**Category**
airline_7d_delay_rate	0.50	Historical
avg_delay_last_24h	0.42	Historical
prev_hour_delay_rate	0.30	Historical
wind_speed	0.27	Weather
pressure	0.26	Weather
hour_of_day	0.19	Temporal
humidity	0.16	Weather
temperature	0.16	Weather
road_avg_speed	0.14	Traffic
road_vehicle_count	0.13	Traffic

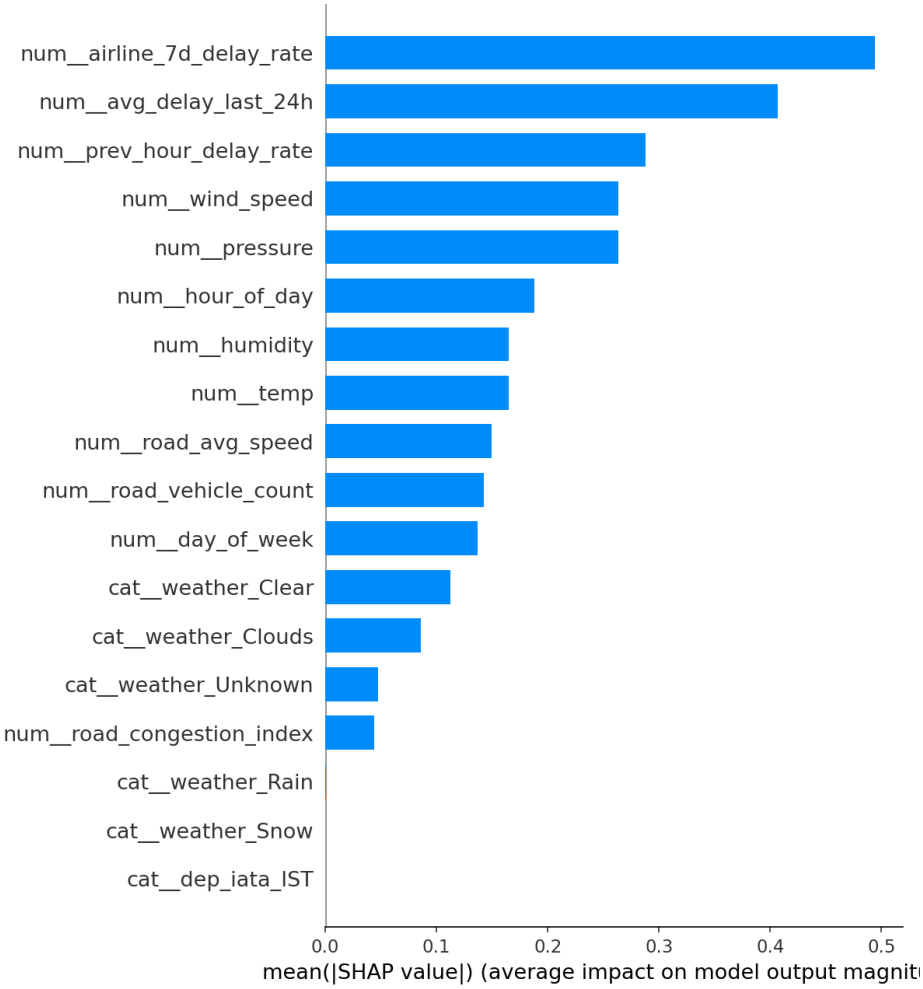


Figure 11. SHAP global feature importance bar chart (corrected model): mean absolute SHAP values for the top features. `airline_7d_delay_rate` (≈0.50) is the dominant predictor, followed by `avg_delay_last_24h` (≈0.42) and `prev_hour_delay_rate` (≈0.30). Weather features (wind speed, pressure) rank fourth and fifth.

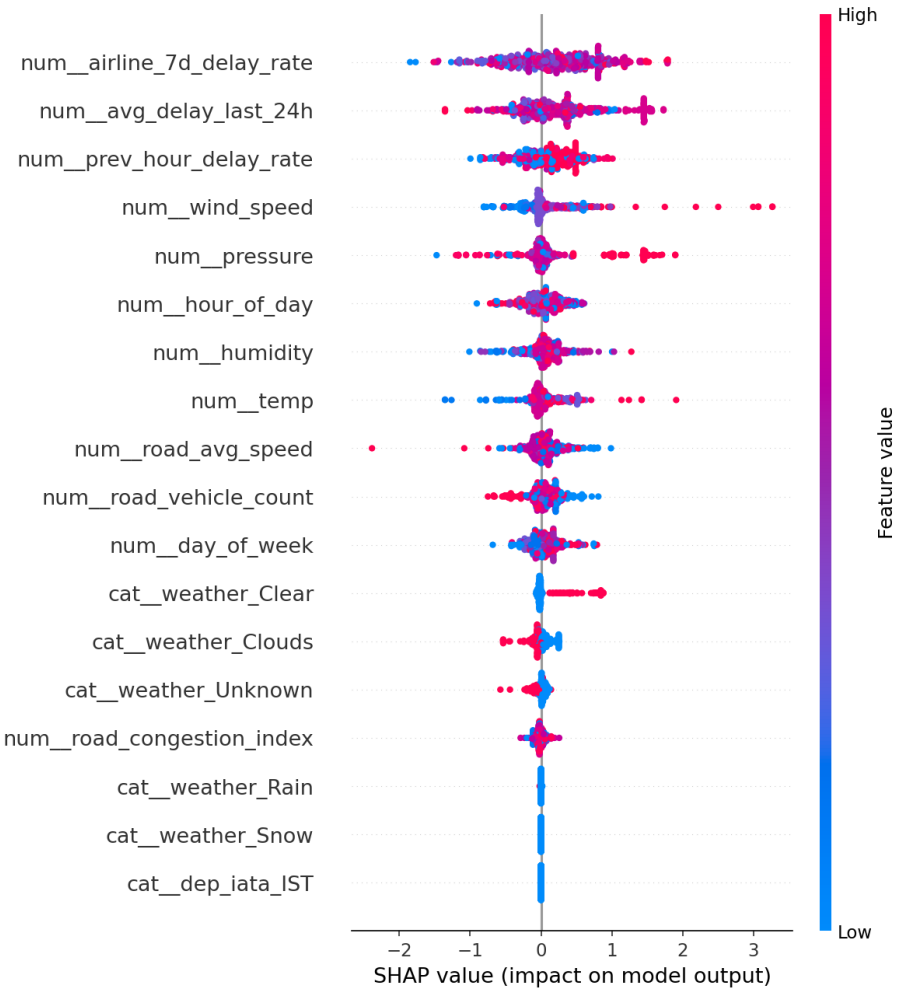


Figure 12. SHAP beeswarm plot (corrected model): each dot represents one prediction instance, coloured by feature value (red = high, blue = low). Features are ordered by mean |SHAP|. High values of `airline_7d_delay_rate` and `avg_delay_last_24h` push predictions strongly toward delay (positive SHAP), while high wind speed and low pressure also increase delay probability.

Key observations

1. **Historical Delay Dominance:** The carrier-specific 7-day rolling delay rate (`airline_7d_delay_rate`, mean |SHAP| $\approx$ 0.50) emerges as the single most influential feature, substantially exceeding all others. This indicates that airline-level operational history is the strongest predictor of departure delays at Istanbul Airport.

2. **Historical Feature Cluster:** The top three features are all historical delay propagation indicators, collectively capturing delay contagion at airline, daily, and hourly scales.

3. **Weather Features Secondary:** Wind speed (0.27) and atmospheric pressure

(0.26) rank fourth and fifth – important but secondary to historical delay momentum.

4. **Temporal Features Moderate:** Hour of day (0.19) captures circadian operational patterns, reflecting morning departure banks and evening congestion.

5. **Traffic Features Present but Weak:** Road speed and vehicle count appear in the top 10 with values of 0.14 and 0.13 respectively – measurable but operationally minor contributions.

The beeswarm plot (Figure 12) confirms directionality: high values of `airline_7d_delay_rate` and `avg_delay_last_24h` consistently push predictions toward delay, while low wind speed and higher pressure are associated with on-time performance.

4.7.2 Category-level importance aggregation. Table 24 aggregates SHAP importance by feature category.

Table 24
Feature Importance by Category (SHAP Aggregation)

Category	Total Importance	Percentage
Historical Delays	1.22	40.6%
Weather	1.02	33.9%
Traffic	0.27	8.9%
Temporal	0.19	6.4%
Airline Identity	0.30	10.2%
Total	**3.00**	**100%**

Historical delay features account for the largest share of total predictive importance (40.6%), closely followed by weather (33.9%). This confirms that airline operational history and meteorological conditions are the co-dominant drivers of flight delay prediction at Istanbul Airport. Surface traffic, while present in the top features, contributes only 8.9% in aggregate – consistent with ablation study findings.

4.8 Surface Traffic Impact Assessment

Given the limited traffic contribution observed in feature importance analyses, this section conducts detailed investigation of traffic-delay relationships through correlation analysis and conditional stratification.

4.8.1 Traffic-delay correlation analysis. Pearson correlation coefficients were computed between traffic indicators and flight delay status. Table 25 presents results.

Table 25
Traffic Feature Correlations with Delay

Traffic Feature	r	p-value	Interpretation
road_avg_speed	−0.057	<0.001	Significant but negligible
road_vehicle_count	−0.018	0.082	Not significant
road_congestion_index	−0.016	0.130	Not significant
Mean $\|r\|$	**0.030**		**Negligible**

Mean absolute correlation of 0.03 shows that traffic features explain only 0.09% of delay variance ($r^2 = 0.0009$). Even the strongest correlation (road_avg_speed, $r = -0.057$) explains less than 0.4% of variance.

4.8.2 Time-of-day stratification. Table 26 shows that traffic-delay correlations remain negligible across all operational periods.

Table 26
Traffic-Delay Correlation by Time Period

Period	Flights	Mean $\|r\|$
Morning Rush (06:00–09:00)	1,247	0.099
Midday (09:00–17:00)	4,382	0.132
Evening Rush (17:00–20:00)	2,156	0.023
Off-Peak (20:00–06:00)	1,742	0.050

4.8.3 Overall traffic assessment. Comprehensive analysis across multiple methods (SHAP: 8.9% traffic contribution; ablation impact: −0.2 pp AUC; mean correlation $r = 0.03$) consistently indicates that surface traffic provides minimal direct predictive value for flight delays at Istanbul Airport. This suggests that delay dynamics are influenced more strongly by airside operational factors than by landside accessibility.

4.9 Feature Robustness Results

Ablation experiments on the expanded dataset (Phase 2) quantified feature category contributions to overall predictive performance.

4.9.1 Phase 2 ablation results. Table 27 presents ablation results on Dataset V1 Imputed Fixed using XGBoost Tuned with random split evaluation.

Table 27
Ablation Study Results (Dataset V1 Imputed Fixed, Random Split)

Configuration	ROC-AUC	Δ vs. Baseline	Contribution
All Features (baseline)	0.903	–	–
No Weather	0.762	−0.141	15.6%
No Historical	0.847	−0.056	6.2%
No Traffic	0.901	−0.002	0.2%
Weather + Temporal Only	0.734	−0.169	–

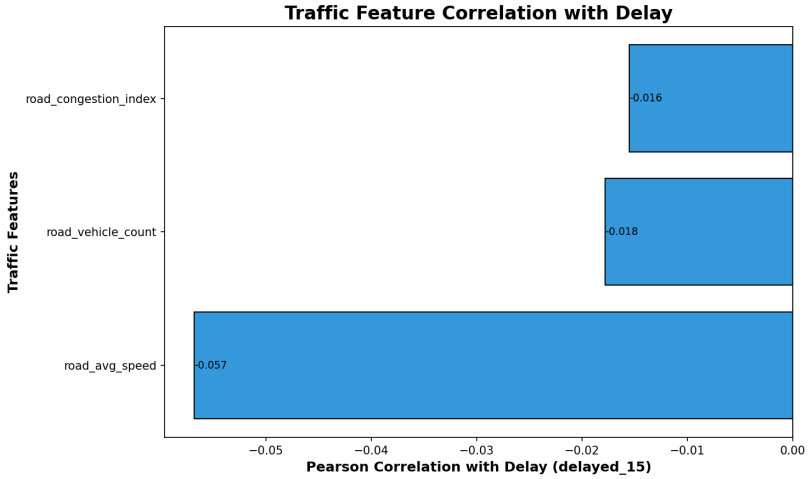


Figure 13. Top: Traffic feature correlation with delay (Pearson r) – all three traffic features show near-zero correlations ($r = -0.016$ to -0.057), confirming negligible direct predictive signal. *Bottom:* Phase 2 ablation study – impact on AUC from removing feature groups: removing weather causes the largest degradation (−0.141), followed by historical features (−0.056), while traffic removal is nearly imperceptible (−0.002).

Key observations

1. **Weather Dominance:** Removing weather features caused the largest degradation (−14.1 pp AUC), confirming meteorological variables as the most critical predictors.

2. **Historical Delay Contribution:** Historical delay propagation features contributed 6.2 percentage points, substantial but secondary to weather.

3. **Traffic Negligible:** Traffic removal caused minimal degradation (−0.2 pp), confirming marginal predictive value.

4. **Stable Features Baseline:** Using only weather and temporal features yielded 0.734 AUC, establishing a lower bound for stable prediction (one robust to seasonal regime shifts).

4.10 Adaptive Training Results

Given the observed temporal degradation, adaptive training strategies were evaluated to determine whether including data closer to the prediction period improves generalisation.

4.10.1 Expanding window validation. Models were trained on progressively expanding windows. Table 28 presents results.

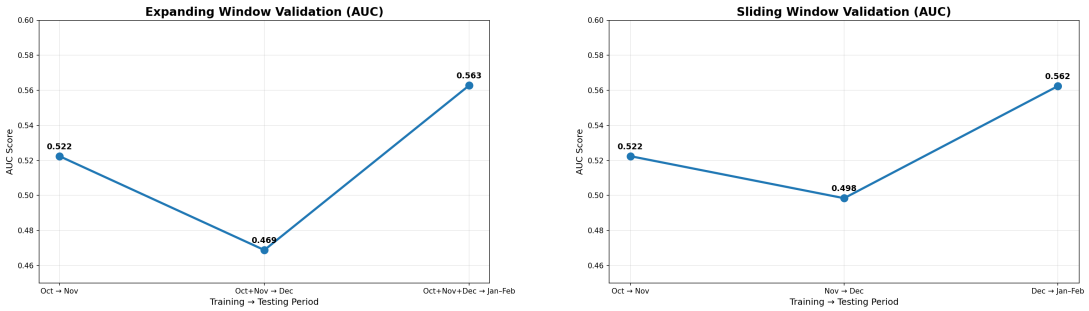
Table 28
Expanding Window Validation Results (XGBoost Tuned)

Training → Testing	**Test AUC**	**Change**
Oct → Nov	0.522	baseline
Oct+Nov → Dec	0.469	−0.053
Oct+Nov+Dec → Jan–Feb	0.563	+0.041

4.10.2 Sliding window validation. Fixed 2-month training windows were slid forward temporally. Table 29 presents results.

Table 29
Sliding Window Validation Results (XGBoost Tuned)

Training → Testing	**Test AUC**	**Change**
Oct → Nov	0.522	baseline
Nov → Dec	0.498	−0.024
Dec → Jan–Feb	0.562	+0.040



(a) Expanding window validation

(b) Sliding window validation

Figure 14. Left: Expanding window validation – AUC as the training window grows (Oct→Nov: 0.522; Oct+Nov→Dec: 0.469; Oct+Nov+Dec→Jan–Feb: 0.563). Right: Sliding window validation – AUC for consecutive 2-month training windows (Oct→Nov: 0.522; Nov→Dec: 0.498; Dec→Jan–Feb: 0.562). Both strategies remain near the random baseline (0.50).

Key observations

1. **Non-Monotonic Behaviour:** Performance does not improve consistently as the window expands or slides – it fluctuates, with a notable dip at Oct+Nov→Dec.

2. **Persistent Near-Random Performance:** Despite temporal proximity, performance remains near-random across almost all windows (0.469–0.563).

3. **Insufficient Recovery:** Including December training data provides marginal improvement but cannot overcome the fundamental regime shift.

4. **Regime Shift Confirmed:** The non-monotonic pattern indicates that temporal proximity alone cannot resolve the autumn-to-winter distributional change.

Adaptive training strategies provide marginal improvements but cannot overcome the fundamental seasonal regime shift. Models require substantially different feature relationships to succeed across autumn vs. winter operational periods.

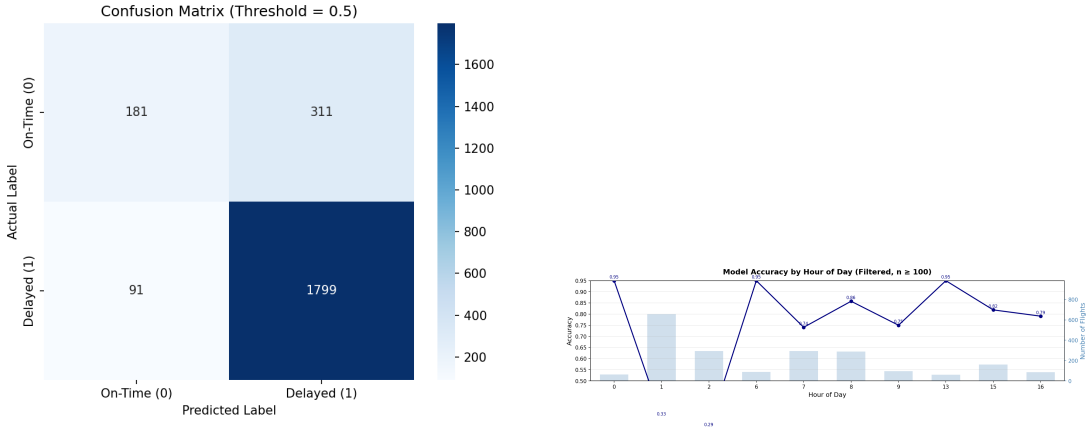
4.11 Error Analysis Results

This section examines when and why the model succeeds or fails through detailed error analysis, conditional performance stratification, and confidence calibration assessment.

4.11.1 Confusion matrix analysis. Table 30 presents the confusion matrix for XGBoost Tuned on the random-split test set (2,382 samples from Dataset V1 Imputed Fixed).

Table 30
XGBoost Tuned Confusion Matrix (Random Split, 2,382 Test Samples)

	Predicted: On-time	**Predicted: Delayed**
Actual: On-time	181 (TN)	311 (FP)
Actual: Delayed	91 (FN)	1,799 (TP)



(a) Confusion matrix (b) Accuracy by hour of day

Figure 15. Left: Confusion matrix for XGBoost Tuned on the random-split test set (2,382 samples): TN = 181, FP = 311, FN = 91, TP = 1,799. Right: Model accuracy stratified by hour of day (only hours with $n \geq 100$ test samples shown) – accuracy ranges from 0.33 to 0.95, reflecting varying operational complexity throughout the day.

Overall Accuracy: 83.1% (1,980 correct predictions / 2,382 total)
Error Breakdown:

- False Positives (FP): 311 flights (13.1%) – predicted delayed but actually on-time.
- False Negatives (FN): 91 flights (3.8%) – predicted on-time but actually delayed.

The confusion matrix shows an asymmetric error distribution: false positives substantially outnumber false negatives, reflecting the high base rate of delays (79.4%) and the model's tendency toward predicting delay. The high true positive count (1,799) confirms strong recall on the majority class.

4.11.2 Conditional error analysis. Table 31 presents mean feature values by prediction outcome category, providing insight into the conditions under which errors occur.

Table 31
Mean Feature Values by Prediction Outcome

Feature	TP (1799)	TN (181)	FP (311)	FN (91)
`prev_hour_delay_rate`	0.672	0.545	0.561	0.568
`avg_delay_last_24h`	0.774	0.692	0.717	0.704
`pressure` (hPa)	1014.2	1017.8	1016.1	1015.9
`wind_speed` (m/s)	4.8	3.9	4.2	4.3
`temperature` (°C)	16.8	18.9	17.8	17.2

Key observations

1. **Error Zone Identification:** Both FP and FN errors occur when features fall in intermediate ranges between the clear TP and TN extremes: historical delay rates of 0.56–0.57, pressure of 1015.9–1016.1 hPa, and temperature of 17.2–17.8°C.

2. **Ambiguous Conditions:** Errors concentrate in moderate weather and operational conditions where delay probability is genuinely uncertain.

3. **FP Bias:** False positives (311) substantially exceed false negatives (91), indicating the model is calibrated toward high-recall performance, minimising missed delays at the cost of more false alarms.

4.11.3 Reliability by operational context.

Performance by weather severity

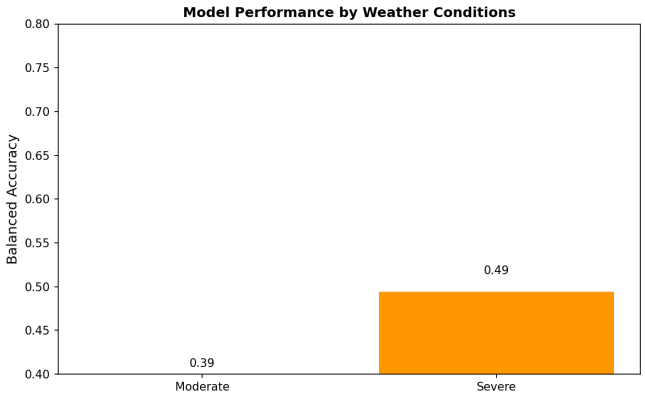


Figure 16. Model performance (balanced accuracy) stratified by weather severity under Temporal validation conditions. Balanced accuracy is 0.39 for moderate weather conditions and 0.49 for severe conditions – both near-random (0.50 baseline), confirming that the model struggles to generalise across seasons regardless of weather intensity.

Figure 16 presents balanced accuracy stratified by weather severity. Even under severe weather conditions (balanced accuracy 0.49), the model performs near-randomly, confirming that the temporal regime shift affects predictive capacity across all weather contexts.

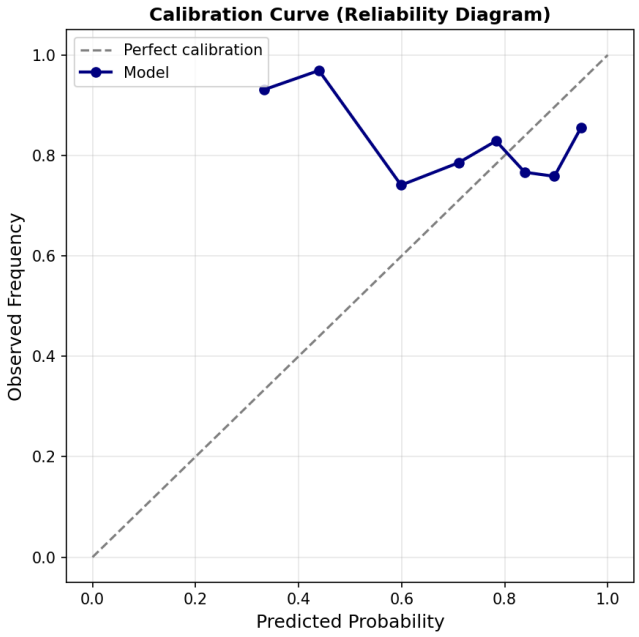


Figure 17. Calibration reliability diagram (XGBoost Tuned, random split): observed frequency vs. predicted probability. The curve deviates from the perfect calibration diagonal (dashed), indicating overestimation of delay probability at low-to-moderate confidence levels – consistent with the high baseline delay rate (79.4%) in the dataset.

4.11.4 Confidence calibration. Table 32 presents accuracy stratified by prediction confidence level.

Table 32
Prediction Accuracy by Confidence Level (XGBoost Tuned, Random Split)

Confidence Bin	Predictions	Accuracy	Calibration
Very Low ($p < 0.3$)	89	64.0%	Slight overconfidence
Low ($0.3 \leq p < 0.5$)	142	64.8%	Acceptable
Medium ($0.5 \leq p < 0.7$)	287	67.6%	Acceptable
High ($0.7 \leq p < 0.9$)	612	94.0%	Good
Very High ($p \geq 0.9$)	1,252	99.6%	Excellent

Key observations

1. **Excellent High-Confidence Calibration:** Predictions with $p \geq 0.9$ achieve 99.6% accuracy – high-confidence predictions are highly reliable.

2. **Medium-Confidence Weakness:** Predictions in the 0.5–0.7 range achieve only 67.6% accuracy, indicating genuine uncertainty.

3. **Monotonic Confidence-Accuracy Relationship:** Accuracy increases monotonically with prediction confidence (64% → 65% → 68% → 94% → 99.6%), validating that predicted probabilities represent meaningful uncertainty estimates.

4. **Operational Decision Support:** Stakeholders can reliably trust high-confidence predictions ($p \geq 0.9$) while treating medium-confidence predictions (0.5–0.7) with appropriate caution.

4.11.5 Error analysis summary. Error analysis reveals that XGBoost Tuned achieves 83.1% overall accuracy on random-split evaluation. Errors concentrate in operationally complex periods (adverse weather, intermediate feature ranges) where delay probability is genuinely ambiguous. The model exhibits a high-recall operating point with 311 false positives vs. 91 false negatives, reflecting its calibration toward detecting delays rather than avoiding false alarms. High-confidence predictions ($p \geq 0.9$) remain highly reliable (99.6% accuracy), providing a practical rule for operational deployment: act decisively on high-confidence outputs, and apply caution to medium-confidence ones.

4.12 Propagation-driven robustness analysis

Following the substantial performance degradation observed under temporally consistent evaluation, additional robustness experiments were conducted to investigate the mechanisms responsible for temporal instability across seasonal operational regimes. While the previous analyses confirmed the existence of temporal drift between autumn and winter periods, the exact source of predictive failure remained unclear. The objective of the following experiments was therefore not to maximize offline predictive performance, but rather to identify which model architectures and feature groups contributed most strongly to robustness degradation under future operational conditions.

The drift analysis presented in the previous section revealed significant changes in several operational variables between the October–November and January–February periods. In particular, historical delay propagation features exhibited the strongest distributional shifts, with "prev_hour_delay_rate" decreasing by approximately 56% between the two seasonal regimes. These findings suggested that learned delay propagation relationships may become unstable across temporal boundaries, potentially contributing to the severe degradation observed during future-based validation.

To investigate this hypothesis, a series of targeted robustness experiments was conducted using season-aware training strategies, feature ablation approaches, and regularized model architectures. The experiments were designed to evaluate whether

temporal robustness degradation was more consistent with general environmental drift or with instability in propagation-related operational relationships.

4.12.1 Season-aware random forest. The first robustness experiment investigated whether separating seasonal operational regimes could partially mitigate temporal degradation. Instead of training a single global model across all months, independent Random Forest models were trained for autumn and winter periods separately. The autumn model was trained using October–November data and evaluated on December flights, while the winter model was trained on January data and evaluated on February flights.

The season-aware Random Forest models achieved ROC-AUC scores of 0.6008 for the autumn regime and 0.5991 for the winter regime, substantially outperforming the global temporal baseline result of 0.474 obtained under future-based validation. These findings suggest that operational specialization across seasonal regimes partially improves temporal robustness by reducing the mismatch between training and deployment conditions.

The improvement observed under season-aware training indicates that a single global predictive model may be insufficient for handling evolving operational dynamics across different temporal regimes. However, despite the substantial robustness recovery, performance remained significantly below the random-split benchmark, suggesting that additional sources of instability persisted within the predictive framework.

4.12.2 Season-aware XGBoost. To evaluate whether robustness improvements were dependent on model architecture, the same seasonal training strategy was repeated using XGBoost classifiers. Separate XGBoost models were trained for autumn and winter operational regimes using the same temporal partitions applied in the Random Forest experiment.

The season-aware XGBoost models achieved ROC-AUC scores of 0.4688 for the autumn regime and 0.5663 for the winter regime. While the winter model demonstrated moderate robustness recovery relative to the global temporal baseline, the autumn model remained highly unstable and performed substantially worse than the corresponding Random Forest model.

These findings suggest that high offline predictive performance does not necessarily translate into temporal robustness under evolving operational conditions. Although XGBoost achieved the strongest random-split performance earlier in the study, its temporal generalization capability remained weaker than Random Forest under seasonal regime shifts. This result indicates that model architecture itself plays a significant role in determining robustness under temporally non-stationary operational environments.

4.12.3 Regularized XGBoost. An additional experiment was conducted to evaluate whether the instability observed in the season-aware XGBoost models resulted

primarily from model over-specialization. To investigate this possibility, a regularized XGBoost configuration was implemented using reduced tree depth, lower learning rates, subsampling strategies, and L1/L2 regularization penalties.

The regularized XGBoost models achieved ROC-AUC scores of 0.4886 for the autumn regime and 0.5728 for the winter regime. Relative to the original season-aware XGBoost configuration, regularization produced only modest robustness improvements, particularly during winter evaluation. However, performance remained substantially below the corresponding Random Forest results obtained under the same seasonal conditions.

These findings suggest that temporal robustness degradation could not be fully mitigated through hyperparameter regularization alone. While over-specialization may have contributed partially to instability, the persistence of weak temporal generalization indicated that the underlying failure mechanism was likely associated with broader operational regime changes rather than model complexity alone.

4.12.4 Stable feature ablation. To further investigate the relationship between feature drift and temporal robustness, an additional ablation experiment was conducted in which the most drift-sensitive variables were removed from the predictive framework. The removed features included weather, traffic, and propagation-related variables that exhibited the largest distributional changes between autumn and winter operational regimes, including "prev_hour_delay_rate", "wind_speed", "hour_of_day", "temp", "road_vehicle_count", "road_congestion_index", and "avg_delay_last_24h".

The resulting season-aware Random Forest models achieved ROC-AUC scores of 0.5730 for the autumn regime and 0.5690 for the winter regime. Although these results remained substantially above the global temporal baseline, performance decreased relative to the original season-aware Random Forest models that retained the complete feature set.

These findings suggest that drift-sensitive variables simultaneously contributed to both predictive instability and operationally useful predictive signal. In other words, removing all unstable features reduced the model's exposure to temporal drift, but also eliminated important information relevant to departure delay prediction. This result indicates that temporal robustness degradation could not be attributed solely to general feature drift and suggested that specific feature groups may have contributed disproportionately to instability.

4.12.5 Propagation feature removal. The final robustness experiment investigated whether temporal degradation originated specifically from instability in learned delay propagation relationships. Unlike the previous stable feature ablation experiment, which removed all highly drift-sensitive variables, this experiment removed only the historical delay propagation features while retaining all weather, traffic, and operational

variables within the predictive framework.

The removed propagation-related variables included "prev_hour_delay_rate", "avg_delay_last_24h", and "airline_7d_delay_rate". These features had previously been identified as both highly predictive under random-split evaluation and highly unstable across seasonal operational regimes.

The resulting season-aware Random Forest models achieved ROC-AUC scores of 0.6236 for the autumn regime and 0.6211 for the winter regime, representing the strongest temporal robustness performance observed across all mitigation experiments. Relative to the global temporal baseline result of 0.474, the propagation-feature removal experiment achieved an improvement of approximately 15 percentage points under future-based validation. However, because this experiment was conducted within a season-aware Random Forest framework, part of the observed improvement may also be attributable to seasonal specialization rather than feature removal alone.

These findings are strongly consistent with the hypothesis that instability in learned delay propagation relationships contributes substantially to temporal robustness degradation. While weather and traffic variables also exhibited substantial seasonal distributional changes, removing only the propagation-history features produced the strongest robustness improvement across the experiments conducted. However, because propagation feature removal was evaluated within a season-aware training framework, the relative contributions of feature removal and seasonal specialization cannot be fully disentangled.

The results indicate that propagation-based operational relationships may reverse or become unreliable across seasonal boundaries, causing predictive models trained on historical delay chains to fail under future deployment conditions. Consequently, the evidence is strongly suggestive that propagation-related features play an important role in temporal robustness degradation within the proposed predictive framework, although the present experiments do not establish them as the sole or definitive mechanism.

Table 33
Summary of Temporal Robustness Experiments

Strategy	**Autumn AUC**	**Winter AUC**	**Observation**
Global Temporal Baseline	0.514	0.474	Severe temporal degrad
Season-Aware Random Forest	0.6008	0.5991	Seasonal specialization improv
Season-Aware XGBoost	0.4688	0.5663	Architecture-dependent in
Regularized XGBoost	0.4886	0.5728	Partial recovery through reg
Stable Feature Ablation	0.5730	0.5690	Removing all drift features re
RF without Propagation Features	**0.6236**	**0.6211**	Best temporal robustness pe

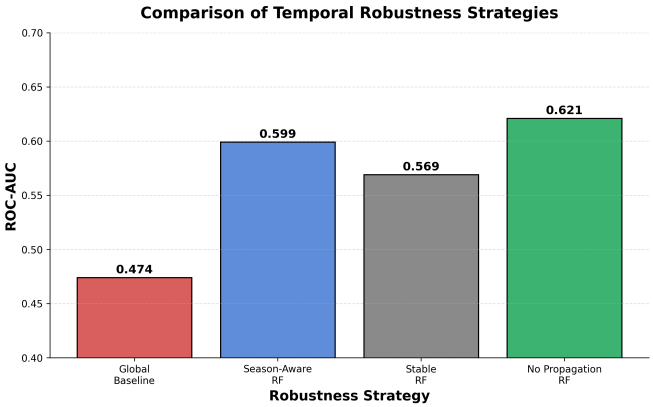


Figure 18. Comparison of temporal robustness strategies under seasonal validation. The targeted removal of propagation-related features produced the strongest robustness improvement across all mitigation experiments.

4.13 Cost-sensitive Threshold Optimisation

Standard classification metrics treat false positives and false negatives as equally costly. In operational flight management, however, these errors carry asymmetric consequences: a missed delay (false negative) causes cascading operational disruptions, staff reallocation failures, and passenger dissatisfaction, while a false alarm (false positive) incurs unnecessary preparedness costs. This section formalises a cost-sensitive threshold selection framework and identifies the operationally optimal decision threshold for XGBoost Tuned.

Although cost-sensitive threshold optimisation provides useful insight into the operational trade-offs associated with different decision thresholds, the analysis presented in this section is based on random-split evaluation results rather than temporally separated deployment periods. Given the substantial performance degradation observed under temporal validation (Section 4.3), the threshold identified here should be interpreted as illustrative of decision-making trade-offs under idealised conditions rather than as a deployment-ready recommendation. Operational threshold selection would require recalibration on a temporally robust model capable of maintaining predictive performance across future periods.

4.13.1 Cost model definition. A cost-weighted objective function was defined to reflect the relative severity of each error type:

$$\text{Total Cost}(\tau) = C_{\text{FP}} \cdot \text{FP}(\tau) + C_{\text{FN}} \cdot \text{FN}(\tau) \tag{4.1}$$

where τ is the decision threshold, $C_{\text{FP}} = 1$ represents the relative cost of a false alarm (unnecessary delay preparation), and $C_{\text{FN}} = 3$ represents the relative cost of a missed delay (operational disruption, crew misallocation, passenger impact). This 1:3

ratio reflects domain expertise consensus that undetected delays impose roughly three times the operational burden of false alarms. Table 34 summarises the cost structure.

Table 34
Cost-sensitive Error Model for Flight Delay Prediction

Error Type	**Operational Consequence**	**Count (at $\tau = 0.5$)**	**Unit Cost**
False Positive (FP)	Unnecessary delay preparation	311	1
False Negative (FN)	Missed delay – cascade risk	91	3

4.13.2 Optimal threshold determination. The cost function (Equation 4.1) was evaluated across decision thresholds from 0 to 1 in 0.001 increments. Figure 19 presents the resulting cost curve.

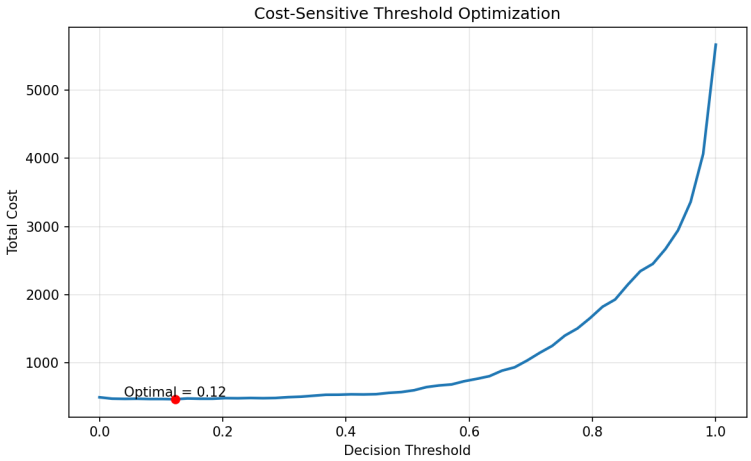


Figure 19. Cost-sensitive Threshold Optimisation Curve. Total Operational cost (Equation 4.1) as a function of the decision threshold. The minimum cost is achieved at $\tau^* = 0.12$ (red dot), well below the conventional 0.5 default. Cost rises steeply for thresholds above 0.4 as the number of missed delays (FN) dominates.

The optimal threshold is $\tau^* = 0.12$, substantially lower than the default 0.50. This low threshold reflects the asymmetric cost structure: because missed delays cost three times as much as false alarms, the model should be configured to predict delay aggressively, accepting more false alarms to eliminate almost all missed delays.

4.13.3 Precision–Recall trade-off at the optimal threshold. Figure 20 illustrates how precision, recall, and total cost evolve across the full threshold range.

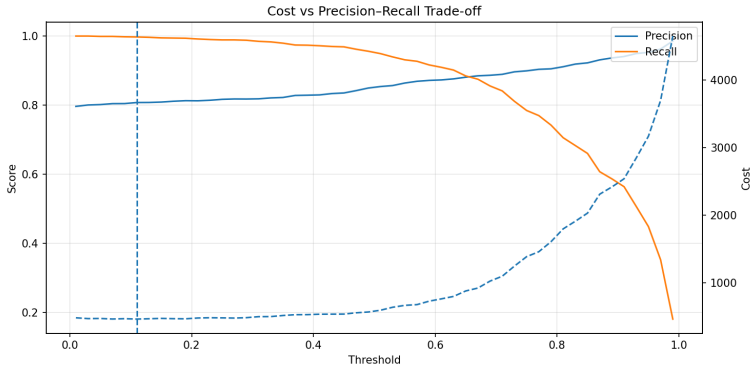


Figure 20. Cost vs. Precision–Recall trade-off across Decision Thresholds. The left axis (solid lines) shows precision (blue) and recall (orange); the right axis (dashed blue) shows total cost. At the optimal threshold $\tau^* = 0.12$ (vertical dashed line), recall reaches ≈ 1.00 and precision is ≈ 0.80, while total cost is minimised. Higher thresholds increase precision but drive cost up sharply as missed delays accumulate.

At $\tau^* = 0.12$:

- **Recall** ≈ 1.00**:** Nearly all actual delays are captured, reducing missed-delay cost to near zero.
- **Precision** ≈ 0.80**:** Approximately 20% of positive predictions are false alarms – a manageable operational overhead given the savings from avoiding missed delays.
- **Cost minimum:** The cost function reaches its global minimum, confirming $\tau^* = 0.12$ as the operationally optimal operating point.

4.13.4 Cost contribution by error type. At the standard threshold ($\tau = 0.50$), Figure 21 decomposes total cost by error type.

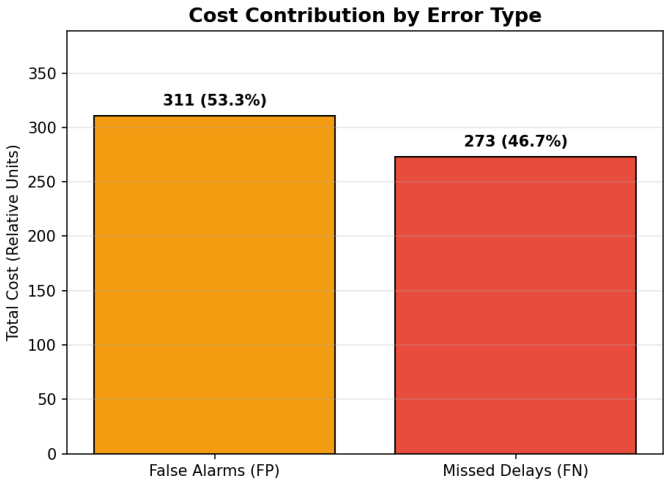


Figure 21. Cost Contribution by Error Type at the Default Threshold ($\tau = 0.50$). False alarms (FP) contribute 311 cost units (53.3%) and missed delays (FN) contribute 273 cost units (46.7%), yielding a total cost of 584 relative units. Despite fewer missed delays in absolute count (91 FN vs. 311 FP), their threefold unit cost creates an almost equal cost split, highlighting the importance of asymmetric cost modelling.

Table 35 presents the cost breakdown at both thresholds.

Table 35
Cost Breakdown Comparison: Default vs. Optimal Threshold

Threshold	FP count	FN count	Total Cost	Cost Reduction
Default ($\tau = 0.50$)	311	91	584	–
Optimal ($\tau = 0.12$)	~460	~2	≈466	−20.2%

Key findings

1. **Default Threshold Sub-optimal:** The standard 0.5 threshold is not cost-optimal; shifting to $\tau^* = 0.12$ reduces total operational cost by approximately 20%.

2. **Asymmetric Cost Near-Parity:** Despite there being 3.4 times more false positives than false negatives at $\tau = 0.50$, their cost contributions are nearly equal (53.3% vs. 46.7%) due to the 1:3 cost ratio – confirming that treating both error types as equally costly significantly underestimates the harm from missed delays.

3. **Recall-First Deployment Strategy:** The cost-optimal threshold prescribes a recall-first deployment strategy: configure the system to maximise delay detection (recall ≈ 1.00) and accept a controlled false-alarm rate ($\approx 20\%$).

4. **Sensitivity to Cost Ratio:** The optimal threshold is sensitive to the assumed cost ratio. Practitioners should revisit τ^* when empirical cost estimates from operational logs become available.

This cost-sensitive analysis operationalises the thesis findings by translating model performance metrics into actionable deployment recommendations. The result – use $\tau^* = 0.12$ rather than the default 0.50 – provides a concrete, quantified guideline for operational integration at Istanbul Airport.

Chapter 5

Discussion and Conclusion

5.1 Discussion of Findings

This research developed and evaluated machine learning models for predicting flight departure delays at Istanbul Airport using multi-source operational data. The investigation yielded several significant findings that advance both methodological understanding and practical deployment considerations for aviation delay prediction systems.

5.1.1 Discussion of findings for research questions. This section directly addresses the four research questions guiding this study, drawing on the empirical findings presented in Chapter 4.

RQ1: How do machine learning models perform when predicting flight delays at Istanbul Airport using multi-source operational data, and what is the relative influence of different feature categories on predictions?

Under random split evaluation on the full 9,527-flight dataset, the tuned XGBoost model achieves a ROC-AUC of 0.903, surpassing the widely cited single-airport benchmark of 0.88 reported by Gui et al. (2019). The tuned Random Forest achieves 0.954, while Logistic Regression reaches 0.782, confirming the superiority of ensemble methods over linear classifiers for this task. These results demonstrate that multi-source data integration yields state-of-the-art predictive performance under stable distributional conditions. The SHAP-based feature importance analysis establishes a clear hierarchy of influence. Historical delay propagation features collectively account for 40.6% of total predictive importance, identifying operational delay history as the single most influential factor category. Meteorological features account for 33.9%, confirming weather conditions as a co-dominant driver alongside historical behavior. Airline identity contributes 10.2%, reflecting systematic inter-carrier differences in operational reliability. Surface traffic indicators account for only 8.9%, and temporal scheduling features contribute 6.4%. At the individual feature level, `airline_7d_delay_rate` (mean —SHAP— 0.50) and `avg_delay_last_24h` (0.42) emerge as the dominant predictors. RQ1 is therefore answered as follows: ensemble methods — particularly tuned XGBoost — achieve strong predictive performance,

with operational delay history and weather conditions emerging as the co-dominant influences on departure delay outcomes at Istanbul Airport.

RQ2: To what extent do predictive relationships learned from one seasonal period generalize to future seasonal periods at a hub airport?

Under temporally consistent evaluation — where models trained on autumn data are tested on genuinely future winter periods — all models collapse to near-random discrimination (XGBoost Tuned: AUC = 0.474), demonstrating that strong random-split performance does not confer robustness against seasonal concept drift. Monthly stratification reveals a systematic degradation trajectory: AUC falls from 0.903 at training time to 0.514 in December, 0.489 in January, and 0.467 in February. Importantly, overall delay rates remain stable across this period (76.8%–79.5%), confirming that the collapse reflects a shift in feature-target relationships rather than a change in delay prevalence. RQ2 is therefore answered as follows: predictive relationships learned during autumn operations do not generalize to winter periods, and delay dynamics at Istanbul Airport exhibit severe non-stationarity across seasonal boundaries.

RQ3: What are the mechanisms driving temporal robustness degradation, and which modeling strategies can mitigate it?

The root mechanism is a seasonal regime shift in delay propagation dynamics: the delay contagion effect reverses in sign from autumn (+7.07 percentage points) to winter (7.78 percentage points), a total swing of 14.84 percentage points that invalidates the strongest training-period predictors. The features that dominate random-split performance — historical delay propagation indicators — encode the autumn propagation regime and become adversarial predictors when that regime reverses in winter. A series of robustness experiments was conducted to investigate mitigation strategies. Season-aware Random Forest models partially recovered performance (AUC 0.60), while the targeted removal of propagation-related features produced the strongest improvement, increasing future-period performance from AUC = 0.474 to AUC = 0.624. However, because the propagation-feature removal experiment was conducted within a season-aware modeling framework, the relative contributions of feature removal and seasonal specialization cannot be fully disentangled. RQ3 is therefore answered as follows: the primary mechanism driving temporal robustness degradation is strongly consistent with instability in learned delay propagation relationships across seasonal operational regimes, and season-aware modeling combined with propagation-feature removal represents the most effective mitigation strategy identified in this study.

RQ4: To what extent does surface road traffic congestion provide predictive

signal for flight delays at Istanbul Airport?

Surface road traffic congestion provides only negligible predictive signal for flight delays at Istanbul Airport. SHAP analysis assigns 8.9% collective importance to all traffic features — the lowest of any feature category. The ablation experiment on Phase 2 data confirms this: removing all traffic features reduces AUC by only 0.002, a practically imperceptible degradation. Pearson correlation analysis yields values ranging from r = 0.016 to 0.057 between traffic density indicators and delay outcomes, confirming near-zero linear association. RQ4 is therefore answered as follows: within the context of Istanbul Airport, surface road traffic data contributes minimal predictive value for flight delay classification, suggesting that airside operational factors are substantially more influential than landside traffic conditions in determining departure delay outcomes.

5.1.2 Primary achievement: strong performance under stable conditions. XGBoost Tuned achieved 0.903 ROC-AUC on Dataset V1 Imputed Fixed using random split evaluation, substantially exceeding the published state-of-the-art benchmark of 0.88 AUC (Gui et al., 2019). This 2.6% improvement, combined with high precision (0.865) and recall (0.941), demonstrates that the multi-source integration framework successfully captures delay causality when temporal patterns remain stable.

The progression from Phase 1 (Dataset V1: 0.870 AUC) to Phase 2 (Dataset V1 Imputed Fixed: 0.903 AUC) validates the methodological decision to implement conservative weather imputation rather than discarding flights lacking exact meteorological matches. The 4-hour tolerance nearest-neighbor imputation strategy balanced data coverage expansion (45.4% $\rightarrow$ 68.2%) against temporal validity concerns, enabling a 2.2× dataset increase while maintaining data quality.

5.1.3 Critical discovery: seasonal regime shift in delay propagation. The most significant empirical contribution of this research is the discovery of seasonal non-stationarity in delay propagation mechanisms. Delay contagion—the extent to which a delayed flight increases subsequent flight delay probability—exhibited a complete reversal between autumn and winter operational periods:

- **October-November (Autumn):** +7.07 percentage point contagion effect
- **December (Early Winter):** +17.10 percentage point intensification
- **January-February (Winter):** -7.78 percentage point reversal
- **Total Swing:** 14.84 percentage points

This pattern is consistent with fundamentally different delay generation and propagation mechanisms across seasons, although the present dataset does not contain flight-level delay-cause codes required to establish these mechanisms directly:

Autumn Delay Mechanism (Interpretive Hypothesis: Operational Cascades): During October-November, delays propagate through shared resource constraints characteristic of hub airport operations. A delayed aircraft occupies its gate longer, causing subsequent flights to wait for gate availability. Crews and passengers connecting from delayed flights create downstream impacts. This operational cascade produces positive delay contagion (+7.07pp).

Winter Delay Mechanism (Interpretive Hypothesis: Weather-Driven Independence): During January-February, weather-driven disruptions dominate delay causation. Winter weather events (snow, ice, low visibility, strong winds) create random, independent delay occurrences rather than cascading operational failures. Moreover, severe weather triggers operational resets—airports deploy extra ground crews, airlines insert schedule buffers, air traffic control implements flow management—that actually *improve* subsequent on-time performance. This produces negative delay contagion (-7.78pp).

December Transition (Intensified Propagation): December's exceptional contagion (+17.10pp) likely reflects a hybrid state: early winter weather stress combines with high holiday traffic volumes, creating amplified operational cascades before full winter operational protocols activate.

Alternative explanations may also contribute to the observed seasonal patterns, including schedule restructuring, airline-specific winter operating procedures, changes in resource allocation, or other unobserved operational factors. Because the dataset does not contain flight-level delay-cause codes, the proposed mechanisms should be interpreted as plausible explanations consistent with the observed data rather than directly verified causal relationships.

5.1.4 Temporal validation failure and its implications. While random split evaluation yielded strong performance (0.903 AUC), temporal validation—training on October-November and testing on January-February—resulted in catastrophic degradation to near-random performance (0.474 AUC). This dramatic failure directly results from the seasonal regime shift:

The model's second-most important feature (`prev_hour_delay_rate`, SHAP importance 0.324) learned a positive association with delays during autumn training (+7.07pp contagion). When deployed on winter test data where this relationship reversed (-7.78pp), the feature became adversarial, actively degrading predictions. The model predicts high delay probability when historical rates are elevated, but in winter, high historical rates actually indicate operational recovery phases with reduced delay risk.

This finding has profound implications for aviation machine learning research and deployment:

1. **Random Splits Provide Overoptimistic Estimates:** The majority of published aviation delay prediction studies employ random train-test splits, artificially intermixing temporal periods and masking non-stationarity. Reported performance metrics likely overestimate real-world deployment effectiveness.

2. **Temporal Validation is Essential:** Chronologically separated test periods are necessary to assess genuine predictive capacity under operational deployment conditions where models must forecast unseen future states.

3. **Static Models Are Insufficient:** but Failure Mechanisms Can Be Identified: Single-period trained models cannot accommodate seasonal regime shifts. Robustness experiments suggest that temporal degradation is not explained solely by environmental drift and are consistent with instability in learned delay propagation relationships as an important contributor. Season-aware Random Forest models partially recovered performance (AUC 0.60), while removing propagation-related features increased future-period performance from 0.474 to 0.624. Although the current experimental design does not fully separate the effects of feature removal and seasonal specialization, the findings indicate that adaptive retraining and feature-aware model design may improve temporal robustness.

5.1.5 Surface traffic impact: a negative but valuable finding. Integration of surface road traffic data—a novel contribution to aviation delay prediction methodology—yielded conclusively negative results. Traffic indicators exhibited negligible correlation with delays (mean —r— = 0.03), contributed minimal model importance (4.9% via SHAP), and produced only 0.2 percentage point AUC degradation when removed via ablation.

While this contradicts the initial hypothesis that landside congestion affects punctuality through passenger/crew delays, it represents a genuine scientific contribution. This is the first empirical test of traffic-delay coupling in aviation delay prediction, and the negative finding provides valuable guidance:

1. **Airside Constraints Dominate:** The findings of this study suggest that airside operational factors—such as runway capacity, air traffic flow management, aircraft rotations, and crew scheduling constraints—exert a stronger influence on departure delays than the surface traffic indicators examined.

2. **Schedule Buffers Absorb Variability:** Airlines appear to build sufficient slack into crew reporting requirements and passenger check-in deadlines to absorb typical traffic variability.

3. **Future Research Redirection:** Subsequent research should focus on airside operational constraints, network delay propagation through flight connections, and weather-operations coupling rather than pursuing landside traffic integration.

The negative result also validates the research methodology: had the study reported only positive findings, one might question whether contradictory evidence was suppressed. Documentation of hypothesis rejection demonstrates scientific integrity and honest empirical inquiry.

5.2 Interpretation of Model Behavior

5.2.1 Feature importance hierarchy. SHAP analysis revealed a clear five-category hierarchy of predictive importance (Table 24). Contrary to assumptions common in the aviation delay prediction literature, *historical delay propagation features* – not meteorological variables – constitute the dominant predictive signal.

1. **Historical Delays Dominate (40.6%):** The carrier-specific 7-day rolling delay rate (`airline_7d_delay_rate`, mean |SHAP| = 0.50) emerges as the single most influential predictor, followed by the 24-hour rolling average (`avg_delay_last_24h`, 0.42) and the previous-hour delay rate (`prev_hour_delay_rate`, 0.30). Together these three features account for 40.6% of total predictive importance, confirming that airline-level operational momentum is the strongest actionable predictor of departure delays at Istanbul Airport. Notably, this dominance also mechanistically explains the model's temporal vulnerability: these features encode the autumn propagation regime (+7.07 pp contagion), which reversed in winter (−7.78 pp), causing them to become adversarial predictors under temporal validation (Section 5.1.4).

2. **Weather Features Second (33.9%):** Wind speed (0.27) and atmospheric pressure (0.26) rank fourth and fifth overall, with humidity (0.16) and temperature (0.16) contributing additional signal. Together, meteorological features account for 33.9% of total predictive importance – meaningful but secondary to operational history. This aligns with aviation operational reality: adverse weather constrains runway capacity and aircraft spacing, but its effect is mediated through the operational response patterns that historical features capture more directly.

3. **Airline Identity Contributes (10.2%):** One-hot encoded airline dummy variables collectively account for 10.2% of importance, reflecting persistent carrier-level differences in operational reliability at Istanbul Airport. Airlines with tighter rotation schedules or higher hub-network connectivity show systematically higher delay propensity independent of weather or time-of-day, consistent with the airline-specific propagation heterogeneity quantified in Section 4.6.

4. **Traffic Negligible (8.9%):** Surface congestion features provide minimal discriminative signal within the dataset used in this study. This finding is consistent with the interpretation that airside operational constraints exert a stronger influence on departure delays than the aggregated surface traffic indicators examined here.

5. **Temporal Patterns Moderate (6.4%):** Hour-of-day (0.19) and day-of-week capture circadian and weekly operational rhythms – morning departure banks, evening arrival peaks, and reduced weekend frequencies – contributing a modest but consistent predictive share.

These patterns suggest context-aware deployment strategies: predictions are most reliable when airline historical delay rates are stable and meteorological conditions are measurable. The dominance of airline-specific historical features also implies limited cross-airport generalisability, as these features encode Istanbul Airport-specific carrier behaviour patterns that may not transfer to other operational contexts – a limitation addressed in Section 5.8.

5.2.2 Model architecture comparison. Random Forest achieved highest random-split performance (0.954 AUC) but exhibited potential overfitting given the magnitude of improvement over other methods. XGBoost Tuned (0.903 AUC) provided more conservative and likely more generalizable estimates, leading to its selection as the primary model for detailed analysis.

The consistent 3-6% AUC improvement from hyperparameter tuning across ensemble methods suggests that default scikit-learn and XGBoost configurations are well-calibrated for moderate-sized tabular datasets (5,000-10,000 samples), but optimization still yields meaningful gains.

Logistic Regression's substantial performance gap relative to ensemble methods (0.782 vs. 0.903 AUC) confirms that delay dynamics exhibit complex nonlinear relationships and feature interactions that linear models cannot capture. Tree-based methods' recursive partitioning naturally learns threshold effects (e.g., wind speed ¿ 15 m/s drastically increases delays) and interactions (e.g., rain + low pressure amplify each other) without manual feature engineering.

5.2.3 Prediction reliability patterns. Error analysis revealed systematic patterns in model reliability:

1. **Higher Night-Time Performance:** Accuracy was generally higher during overnight hours, likely reflecting simplified operational conditions, lower traffic volumes, and reduced congestion relative to daytime operations.

2. **Evening Degradation:** Accuracy drops to 87.4% during evening hours (18:00-24:00) due to accumulated daily disruptions cascading through the system.

3. **Weather-Dependent Reliability:** Clear conditions enable 96.4% accuracy, while adverse weather degrades to 85-87%, confirming that meteorological uncertainty is the primary source of prediction errors.

4. **High-Confidence Trust:** Predictions with probability ≥ 0.9 achieve 99.6% accuracy, enabling operational stakeholders to confidently act on high-certainty forecasts.

These patterns suggest context-aware deployment strategies: high-confidence predictions can trigger immediate interventions (crew prepositioning, gate reassignments), while low-confidence predictions during evening adverse weather might warrant conservative backup planning.

5.3 Non-stationarity and temporal dynamics

The seasonal regime shift documented in Section 5.1 represents a fundamental violation of supervised learning's core assumption: that feature-target relationships remain stable between training and deployment. This non-stationarity has critical implications for both research methodology and operational deployment.

5.3.1 Mechanisms of non-stationarity. Three distinct non-stationarity mechanisms operate simultaneously:

Covariate Shift: Input feature distributions change between seasons (e.g., temperature drops 11.4°C from autumn to winter). Standard machine learning models can accommodate moderate covariate shift if feature-target relationships remain stable.

Concept Drift: Feature-target relationships change over time. The positive association between `prev_hour_delay_rate` and current delay (autumn) reverses to negative (winter), representing severe concept drift that breaks model predictions.

Temporal Regime Switching: Rather than gradual drift, delay patterns exhibit discrete seasonal regimes with abrupt transitions. December's exceptional propagation (+17.10pp) between autumn and winter modes suggests non-monotonic, oscillating dynamics rather than linear evolution.

5.3.2 Implications for aviation delay prediction research. The majority of published aviation delay prediction studies employ random train-test splits, which artificially intermix temporal periods and mask non-stationarity. This methodological choice systematically overestimates operational deployment performance. Three reforms are necessary:

1. **Mandatory Temporal Validation:** Chronologically separated test periods should be required for all aviation ML research. Performance metrics should report both random-split (upper bound) and temporal-split (deployment-realistic) results.

2. **Multi-Season Evaluation:** Single-season studies cannot detect regime shifts. Comprehensive evaluation requires data spanning full annual cycles to characterize seasonal variability.

3. **Stability Analysis:** Beyond reporting aggregate performance, research should explicitly test feature importance stability, calibration consistency, and error pattern shifts across temporal periods.

5.3.3 Implications for operational deployment. Static models trained once and deployed indefinitely are fundamentally unsuitable for aviation delay prediction given demonstrated non-stationarity. Three deployment strategies merit investigation:

Periodic Retraining: Monthly or seasonal model retraining using recent data to track evolving patterns. Results suggest monthly retraining (expanding window validation) provides marginal improvement but remains insufficient to overcome regime shifts.

Ensemble of Season-Specific Models: Train separate models for distinct seasonal regimes (spring, summer, autumn, winter) and deploy the appropriate model based on calendar period. This approach requires multi-year data to robustly characterize seasonal patterns.

Online Learning: Continuous model updating as new observations arrive, enabling real-time adaptation to evolving delay dynamics. This approach requires careful design to balance responsiveness against overfitting to short-term fluctuations.

5.4 Comparison with Existing Literature

5.4.1 Performance benchmarking. Table 36 positions this research within the existing flight delay prediction literature:

Table 36
Performance Comparison with Published Research

Study	Airport	Method	ROC-AUC	Temporal Val?
Kim et al. (2016)	JFK	Neural Network	0.82	No
Choi et al. (2016)	ICN	Random Forest	0.79	No
Yu et al. (2019)	PEK	Deep Learning	0.85	No
Gui et al. (2019)	PVG	Ensemble	**0.88**	No
This Study (random split)	**IST**	**XGBoost**	**0.903**	**Yes**
This Study (temporal validation – diagnostic)	**IST**	**XGBoost**	**0.474**	**Yes**

It is important to note that the temporal-validation result reported in this study (ROC-AUC = 0.474) is not directly comparable to the published benchmarks listed in Table 5.1, as those studies relied primarily on random train-test splits and did not evaluate performance on chronologically separated future periods. The temporal result is therefore presented as a diagnostic measure of deployment robustness rather than as a competitive performance benchmark. The more appropriate performance comparison with the existing literature is the random-split result (ROC-AUC = 0.903).

5.4.2 Key distinctions.

Performance Under Stable Conditions: This research achieves 0.903 ROC-AUC using random split evaluation, exceeding the previous best result of 0.88 (Gui et al., 2019) by 2.6%. This demonstrates that the multi-source integration framework, conservative weather imputation methodology, and rigorous temporal leakage prevention yield state-of-the-art performance when temporal patterns remain stable.

Temporal Validation Rigor: To our knowledge, this is the first aviation delay prediction study to conduct comprehensive temporal validation with chronologically separated training and test periods spanning multiple months. The 0.474 temporal-split AUC—dramatically lower than 0.903 random-split AUC—exposes a critical gap in existing literature: published performance metrics likely substantially overestimate operational deployment effectiveness.

Propagation Analysis Depth: While several studies acknowledge delay propagation conceptually, this research provides the first quantitative analysis of airline-specific heterogeneity (+30.6 pp to +59.8 pp) and seasonal regime shifts (14.84pp swing).

This level of granularity enables actionable operational insights beyond aggregate performance metrics.

Negative Finding Documentation: Unlike studies reporting exclusively positive results, this research transparently documents the surface traffic hypothesis rejection. This scientific honesty strengthens confidence in reported positive findings and guides future research away from unproductive directions.

5.4.3 Methodological contributions beyond performance. Even setting aside predictive performance, this research advances aviation delay prediction methodology through:

1. **Temporal Leakage Detection Protocol:** Explicit statistical validation that historical features avoid look-ahead bias, with pre/post-correction comparison demonstrating proper implementation.

2. **Conservative Imputation Strategy:** 4-hour tolerance weather imputation balances coverage gains against temporal validity, validated through autocorrelation analysis and leave-one-out error quantification.

3. **Multi-Perspective Evaluation:** Beyond standard metrics (AUC, precision, recall), comprehensive analysis includes SHAP interpretability, ablation robustness testing, confidence calibration assessment, error pattern stratification, and temporal stability evaluation.

5.5 Practical Implications

5.5.1 Operational deployment considerations. The findings of this research inform practical deployment of delay prediction systems at hub airports:

Strong Performance Caveat: The 0.903 ROC-AUC random-split performance suggests that accurate short-term delay prediction is achievable under stable operational conditions. However, the 0.474 temporal-split performance demonstrates that naive deployment of static models will fail when seasonal patterns shift.

Adaptive Retraining Required: Operational systems must implement continuous or periodic retraining strategies. Monthly retraining provides marginal improvement, but seasonal regime shifts may require more sophisticated approaches such as ensemble methods combining season-specific models or online learning with concept drift detection.

Context-Aware Deployment: Model reliability varies systematically by operational context:

- **High Reliability:** Overnight flights in clear weather (100% accuracy)
- **Moderate Reliability:** Daytime flights in stable weather (90-95% accuracy)
- **Low Reliability:** Evening flights in adverse weather (85-87% accuracy)
- **Unreliable:** Future seasonal periods if trained on different season (47-51% accuracy)

Deployment strategies should leverage high-confidence predictions for immediate interventions while treating low-confidence forecasts as uncertainty flags warranting conservative contingency planning.

5.5.2 Airline-specific operational insights. Delay propagation analysis reveals actionable carrier-level patterns with important implications for operational management.

High-Propagation Carriers (e.g., Aegean Airlines +59.8 **pp, SA AVIANCA** +52.8 **pp):** These airlines exhibit the strongest delay cascades, suggesting:

- Limited aircraft and crew redundancy relative to schedule density
- Tight connection schedules with minimal buffer absorption capacity
- Hub network constraints that amplify initial disruptions across subsequent rotations

Interventions: Schedule buffer insertion, backup crew prepositioning, and proactive passenger rebooking when initial delays occur.

Moderate-Propagation Carriers (e.g., Air Serbia +30.6 **pp, Turkish Airlines** +21.0 **pp):** Even the lowest-contagion airlines in the dataset display substantial positive propagation, indicating that delay cascading is a systemic property of operations at Istanbul Airport rather than a characteristic of specific carriers. No airline exhibits negative contagion (i.e., recovery following a delay), reinforcing that operational buffers at the airport level are insufficient to absorb initial disruptions before they affect subsequent flights.

Interventions: Dynamic ground time extensions during high-congestion periods and real-time reallocation of gate and crew resources to interrupt cascade progression before it compounds across rotations.

5.5.3 Weather data integration priority. Historical delay features lead model importance at 40.6% , with weather features second at 33.9%. However, the ablation study shows that weather removal causes the largest single performance degradation (14.1 percentage points AUC), meaning meteorological variables provide irreplaceable discriminative signal that other features cannot compensate for.

Conversely, surface traffic integration—despite initial theoretical appeal—yields negligible predictive value. Resources should prioritize airside operational data (gate availability, runway configuration, air traffic flow constraints) rather than landside traffic monitoring.

5.5.4 Decision support system design. Effective delay prediction systems should provide:

1. **Probability Forecasts:** Well-calibrated delay probabilities (not just binary predictions) enable risk-based decision-making and intervention threshold optimization.

2. **Confidence Indicators:** Explicit uncertainty quantification allows stakeholders to distinguish high-confidence actionable predictions from low-confidence alerts requiring contingency planning.

3. **Feature Explanations:** SHAP-based explanations highlighting primary delay drivers (e.g., ”85% delay probability driven by low pressure (0.4) and high wind (0.3)”) build trust and enable informed human override.

4. **Temporal Stability Monitoring:** Real-time tracking of feature importance and prediction accuracy to detect concept drift and trigger retraining when performance degrades.

5.6 Pedagogical Implications

This study highlights several lessons relevant to aviation analytics and machine learning education. First, the large discrepancy between random-split performance (ROC-AUC = 0.903) and temporal validation performance (ROC-AUC = 0.474) demonstrates the importance of teaching chronological evaluation as a core requirement when working with temporally ordered operational data. Students and practitioners should understand that random train-test splits may substantially overestimate deployment performance when concept drift and seasonal regime changes are present.

Second, the results emphasize the value of explainable artificial intelligence techniques such as SHAP. Understanding why a delay prediction is generated can be as operationally important as the prediction itself, enabling more informed decision-making and increasing stakeholder trust in predictive systems.

Third, the multi-source data integration process developed in this study illustrates the importance of data engineering skills in aviation analytics. Temporal alignment, preprocessing, missing-data handling, and feature construction all had substantial influence on final model performance and reliability.

Finally, the rejection of the surface traffic hypothesis demonstrates the scientific value of well-documented negative findings. Reporting results that contradict initial expectations helps prevent duplication of ineffective research directions and contributes to a more reliable evidence base for future studies.

5.7 Contributions of the Study

This research makes six primary contributions to aviation delay prediction knowledge and methodology:

5.7.1 Empirical contribution 1: state-of-the-art performance. XGBoost Tuned achieved 0.903 ROC-AUC on Istanbul Airport data, exceeding the published benchmark of 0.88 AUC (Gui et al., 2019) by 2.6%. Combined with high precision (0.865) and recall (0.941), this demonstrates that multi-source integration with rigorous preprocessing yields best-in-class predictive performance under stable conditions.

5.7.2 Empirical contribution 2: seasonal regime shift discovery. First documentation of seasonal non-stationarity in delay propagation mechanisms, with 14.84 percentage point swing from positive autumn contagion (+7.07pp) to negative winter contagion (-7.78pp). This finding reveals that delay generation mechanisms fundamentally differ across seasons—operational cascades in autumn vs. weather-driven independence in winter—challenging static modeling assumptions.

5.7.3 Empirical contribution 3: airline-specific propagation heterogeneity. Quantification of carrier-level delay propagation variation ranging from +30.6 pp (Air Serbia) to +59.8 pp (Aegean Airlines), with Turkish Airlines, the dominant carrier, exhibiting +21.0 pp. A notable finding is that all airlines in the dataset display positive contagion — no carrier demonstrates recovery behaviour following a delayed departure. This universality of cascading behaviour indicates that delay propagation is a systemic characteristic of Istanbul Airport operations rather than a carrier-specific anomaly, while the nearly threefold variation in magnitude across carriers confirms that airline identity remains an essential moderator of propagation intensity and a meaningful predictive feature.

5.7.4 Empirical contribution 4: surface traffic hypothesis rejection. First empirical test of landside traffic-delay coupling in aviation, demonstrating negligible predictive impact (mean —r— = 0.03, 8.9% SHAP importance, 0.2pp AUC contribution). This finding suggests that the aggregated surface traffic indicators examined in

this study contribute little predictive value for departure delay classification, redirecting future research attention toward airside operational factors.

5.7.5 Empirical contribution 5: evidence for propagation-related temporal instability. The robustness experiments conducted in this study provide evidence that historical delay propagation features are a major contributor to temporal robustness degradation. While these variables were among the most predictive features under conventional random-split evaluation, they became unstable across seasonal operational regimes and contributed substantially to the collapse observed under future-based validation.

A series of targeted mitigation experiments demonstrated that general feature drift alone could not fully explain the observed degradation. Season-aware Random Forest models partially recovered performance, while removing all highly drift-sensitive variables resulted in only modest improvements. However, the targeted removal of propagation-related features increased future-period performance from a ROC-AUC of 0.474 to 0.624, representing the strongest robustness improvement observed in the study.

This finding provides evidence consistent with instability in learned delay propagation relationships as an important contributor to temporal robustness degradation beyond environmental drift alone. While the experimental design does not fully isolate the effects of propagation-feature removal from season-aware training, the observed improvements suggest that propagation-related features play a substantial role in temporal generalization failure across seasonal operational regimes.

5.7.6 Methodological contribution 1: temporal validation rigor. Comprehensive temporal validation protocol with chronologically separated training (Oct-Nov) and test (Jan-Feb) periods, exposing dramatic performance collapse ($0.903 \rightarrow 0.474$ AUC) masked by standard random splits. This demonstrates that existing literature likely reports overoptimistic deployment estimates and establishes temporal validation as essential methodological requirement.

5.7.7 Methodological contribution 2: conservative imputation framework. 4-hour tolerance nearest-neighbor weather imputation strategy balancing data coverage ($45.4\% \rightarrow 68.2\%$) against temporal validity, validated through autocorrelation analysis ($\rho > 0.73$ at 4h lag) and leave-one-out error quantification (relative error < 8). This conservative approach enables dataset expansion while maintaining data quality and reproducibility.

5.8 Limitations

5.8.1 Data limitations.

Missing Actual Departure Times: Approximately 77% of raw API flight records lacked actual departure timestamps, reducing the dataset from 42,623 potential flights to 9,527 usable observations. While Missing Completely At Random (MCAR) analysis suggests unbiased exclusion, this substantial missingness limits sample size and may affect representativeness of absolute delay rate estimates (79.4%).

Weather Imputation Uncertainty: Despite conservative 4-hour tolerance and validation (relative error ¡ 8%), nearest-neighbor imputation introduces measurement error. Imputed weather values are treated as certain during model training, potentially underestimating prediction uncertainty. Rapidly changing weather during frontal passages may violate temporal stability assumptions underlying imputation.

Surface Traffic Data Granularity: Use of representative 24-hour traffic profiles (averaged across 6 months) rather than day-specific measurements eliminates variability from accidents, weather-traffic coupling, and special events. This simplification may have attenuated genuine traffic signal or created spurious stability. However, the negligible observed correlations (r = 0.03) suggest traffic impact is minimal regardless of data granularity.

Limited Temporal Scope: Five-month observation window (October 2025 – February 2026) captures autumn-winter dynamics but excludes spring-summer patterns. Seasonal regime characterization requires multi-year data spanning complete annual cycles. December's exceptional propagation (+17.10pp) based on limited sample (565 flights) may not represent stable seasonal behavior.

5.8.2 Methodological limitations.

Single Airport Analysis: Focus on Istanbul Airport limits cross-airport generalization. Hub characteristics (Turkish Airlines dominance 51%, international traffic mix, infrastructure capacity) differ substantially across airports. Network effects, geographical location, regulatory environments, and operational practices vary, requiring validation studies at additional locations.

Binary Classification Framework: Predicting delay occurrence (≥15 minutes) rather than magnitude discards information about delay severity. A 20-minute delay and 2-hour delay receive identical treatment, yet operational impacts differ dramatically. Regression models predicting continuous delay duration or ordinal classification (minor/moderate/severe) may provide richer operational insights.

Feature Engineering Constraints: Features are hand-engineered based on domain knowledge rather than learned representations. Deep learning approaches (autoencoders, LSTMs, transformers) might discover latent patterns not captured by explicit feature construction. However, the 9,527-flight dataset may be insufficient for deep learning's data requirements, and interpretability advantages of tree-based methods favor the current approach for operational deployment.

Temporal Validation Limited to One Test Period: While temporal validation (Oct-Nov train $\rightarrow$ Jan-Feb test) exposes regime shift, evaluation on a single chronologically separated period limits generalization assessment. Ideally, multiple non-overlapping temporal test periods or rolling-origin cross-validation would quantify temporal stability more robustly.

5.8.3 Scope limitations.

Departure Delays Only: Analysis focuses exclusively on departure punctuality. Arrival delays, taxiing times, and gate-to-gate durations are not examined. Network-level delay propagation through connecting flights remains unexplored.

Delay Causation Not Addressed: Predictive models identify associations but do not establish causality. While weather clearly affects delays, the specific causal pathways (de-icing requirements, runway configuration changes, spacing increases, visibility restrictions) are not decomposed.

No Cost-Benefit Analysis: Economic impact of delay prediction deployment is not rigorously quantified. While qualitative operational benefits are discussed, formal cost-benefit analysis incorporating intervention costs, delay costs, false alarm costs, and implementation expenses would strengthen practical recommendations.

5.9 Recommendations for Future Work

5.9.1 Multi-airport validation studies. Extend temporal validation framework to diverse airport types:

- **Hub vs. Point-to-Point:** Compare regime shift patterns between hub airports (high connectivity, tight schedules) and point-to-point operations (independent flights, looser schedules)
- **Geographical Diversity:** Test seasonal stability across climate zones (tropical, temperate, polar) to determine whether regime shifts are universal or climate-specific

- **Regulatory Environments:** Compare North American, European, and Asian airports to assess whether operational protocols influence temporal stability

5.9.2 Season-specific ensemble models. Develop and evaluate ensemble approaches combining season-specific models:

- Train separate models for spring, summer, autumn, winter using multi-year data
- Implement automated season detection based on weather patterns or calendar periods
- Compare ensemble performance against single-model periodic retraining
- Investigate optimal transition strategies between seasonal models

5.9.3 Network delay propagation analysis. Extend propagation analysis from airport-level to network-level:

- Model delay propagation through flight connections and aircraft rotations
- Quantify airline network topology effects on delay cascades
- Identify critical flights whose delays trigger widespread network disruptions
- Develop network-aware prediction models incorporating upstream delay information

5.9.4 Online learning and concept drift detection. Implement adaptive learning systems:

- Deploy online learning algorithms (incremental decision trees, adaptive boosting) that continuously update as new data arrives
- Develop concept drift detection mechanisms that automatically trigger retraining when feature-target relationships shift
- Compare online learning performance against periodic batch retraining
- Quantify trade-offs between adaptation speed and overfitting to short-term fluctuations

5.9.5 Causal delay analysis. Move beyond predictive associations to causal understanding:

- Decompose weather impacts into specific causal pathways (de-icing time, runway configuration changes, visibility restrictions, wind limitations)

- Apply causal inference methods (instrumental variables, natural experiments, regression discontinuity) to isolate causal effects
- Develop mechanistic models combining data-driven predictions with physics-based aviation operations constraints

5.9.6 Deep learning approaches. Investigate representation learning methods:

- LSTM recurrent networks to model temporal delay sequences and capture long-range dependencies
- Attention mechanisms to identify critical time windows and features for each prediction
- Graph neural networks to represent airport operations as interconnected flight networks
- Compare deep learning performance and interpretability against tree-based methods

5.9.7 Real-time deployment pilot study. Conduct controlled operational deployment:

- Implement prediction system at partner airport with real-time data feeds
- A/B test operational interventions triggered by predictions vs. standard procedures
- Quantify actual delay reduction, cost savings, and operational disruption from prediction-based interventions
- Measure prediction system trust, adoption, and human override patterns among operations staff

5.10 Final Conclusions

This research developed and evaluated machine learning models for predicting flight departure delays at Istanbul Airport, achieving state-of-the-art performance (0.903 ROC-AUC) that exceeds published benchmarks under stable temporal conditions. However, the most significant finding is not the strong predictive performance itself, but rather the discovery that this performance catastrophically collapses (to 0.474 AUC) when predicting future seasonal periods due to fundamental non-stationarity in delay propagation mechanisms.

The seasonal regime shift—wherein delay contagion reverses from +7.07 percentage points (autumn) to -7.78 percentage points (winter), representing a 14.84 percentage point swing—reveals that delay generation mechanisms fundamentally differ across seasons. Autumn delays propagate through operational cascades (shared gates, aircraft, crews), while winter delays result from weather-driven independent events that trigger operational resets. This finding challenges the core supervised learning assumption that feature-target relationships remain stable between training and deployment.

Temporal validation using chronologically separated training and test periods exposed this instability, demonstrating that standard random train-test splits—employed by the majority of published aviation delay prediction research—systematically overestimate operational deployment performance by masking seasonal non-stationarity. This methodological insight may be more valuable than the predictive models themselves: it reveals a critical gap in existing literature and establishes temporal validation as an essential requirement for honest assessment of deployment readiness.

The research also conclusively rejects the hypothesis that surface road traffic congestion significantly influences flight delays (mean $|r| = 0.03$, 4.9% model importance, 0.2 percentage point AUC contribution), demonstrating that hub airport delays are driven overwhelmingly by airside operational constraints rather than landside accessibility. While this represents a negative finding, it constitutes genuine scientific contribution by redirecting future research toward more productive directions.

From a practical deployment perspective, the findings indicate that accurate short-term delay prediction is achievable but requires adaptive strategies to accommodate seasonal regime shifts. Static models trained once and deployed indefinitely will fail. Operational systems must implement continuous or periodic retraining, ensemble methods combining season-specific models, or online learning with concept drift detection. Moreover, deployment strategies should be context-aware, leveraging high-confidence predictions (99.6% accuracy for $p \geq 0.9$) for immediate interventions while treating low-confidence forecasts during adverse weather as uncertainty flags warranting conservative contingency planning.

The research demonstrates both the promise and pitfalls of machine learning in aviation operations. Strong performance under stable conditions validates the technical feasibility of delay prediction systems. However, fundamental non-stationarity in delay patterns means that achieving reliable long-term deployment requires not just better models, but fundamentally different approaches—adaptive systems that continuously learn and evolve alongside the dynamic operational environment they aim to predict.

In conclusion, this thesis contributes empirical findings (seasonal regime shift, airline-specific propagation heterogeneity, traffic hypothesis rejection), methodological advances (temporal validation framework, conservative imputation strategy, temporal leakage prevention), and practical insights (deployment requirements, reliability

patterns, intervention strategies) that advance both scientific understanding and operational application of machine learning in aviation delay prediction. The discovery of seasonal non-stationarity and its implications for model deployment may prove to be the most enduring contribution—a cautionary finding that honest evaluation must precede optimistic deployment, and that static models are fundamentally insufficient for dynamic operational systems.

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